

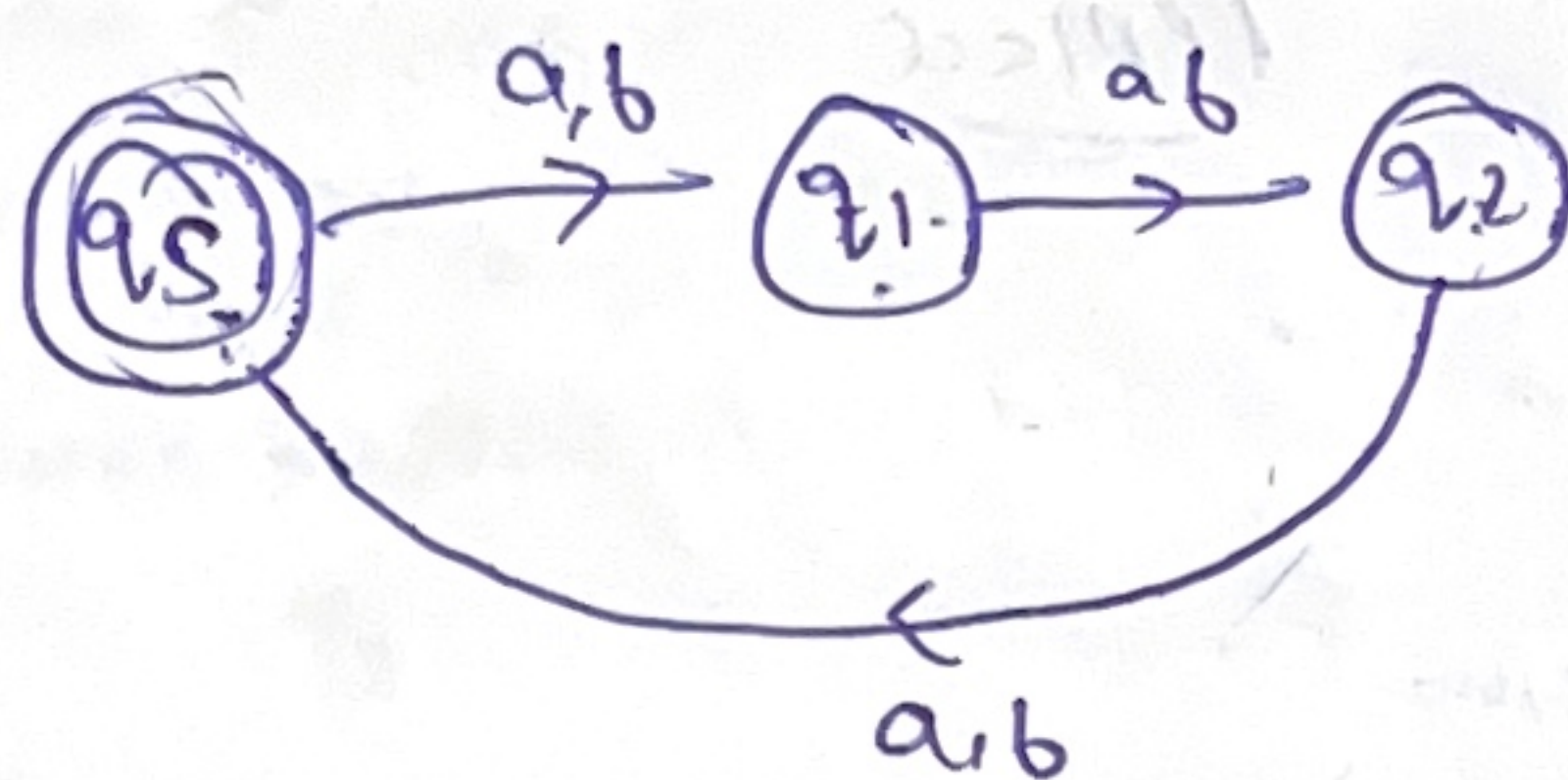
Q1) Show that the following language is Regular.

$$L = \{a^n \mid n \geq 4\}$$

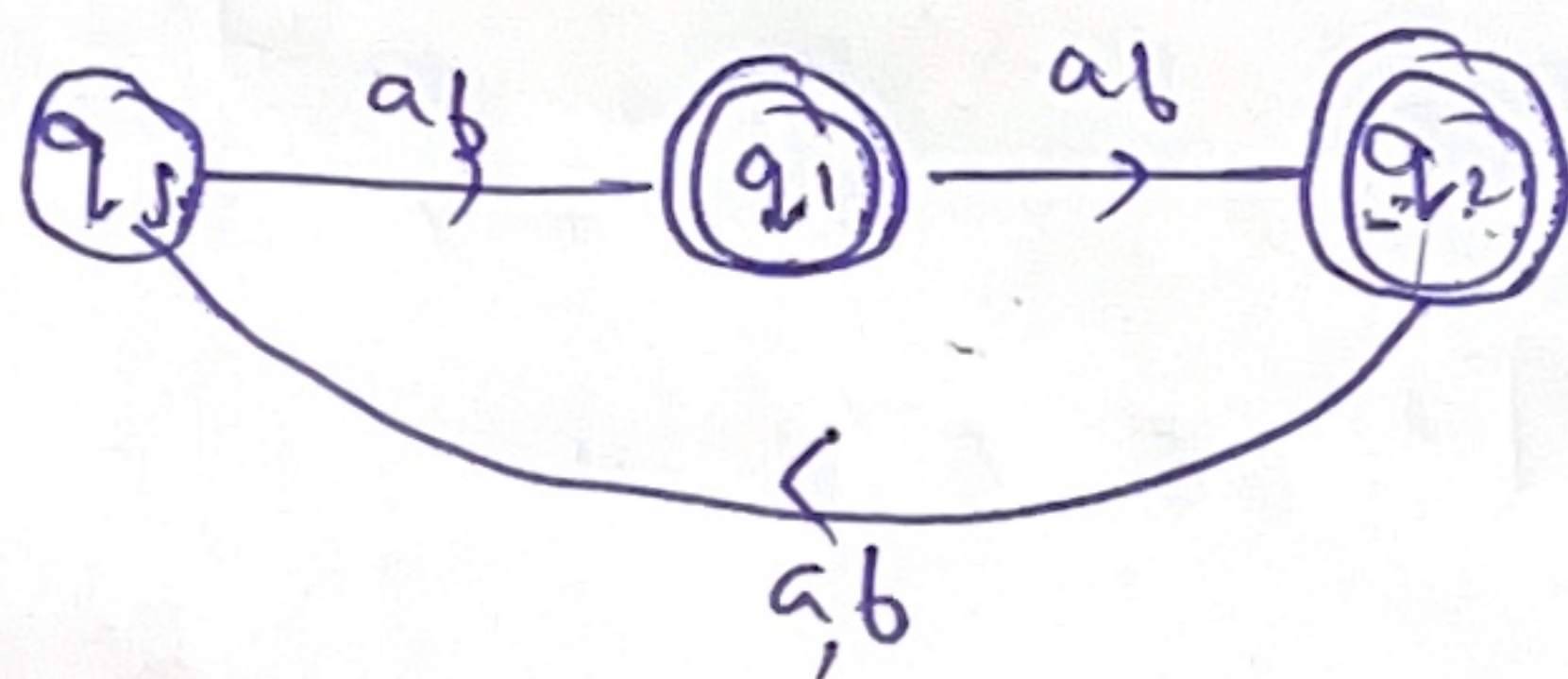
$$L = \{aaaa, aaaaa, \dots\}$$

this language can be represented by Regular expression $aaaaa^*$

Q2)
a) $\{L: w \mid |w| \bmod 3 = 0\}$



b) $\{L: w \mid |w| \bmod 3 \neq 0\}$



Context free grammar

$$A \rightarrow \alpha; \text{ where } \alpha \in (V \cup \Sigma)^*$$

Ex: $P_1: S \rightarrow aSa$

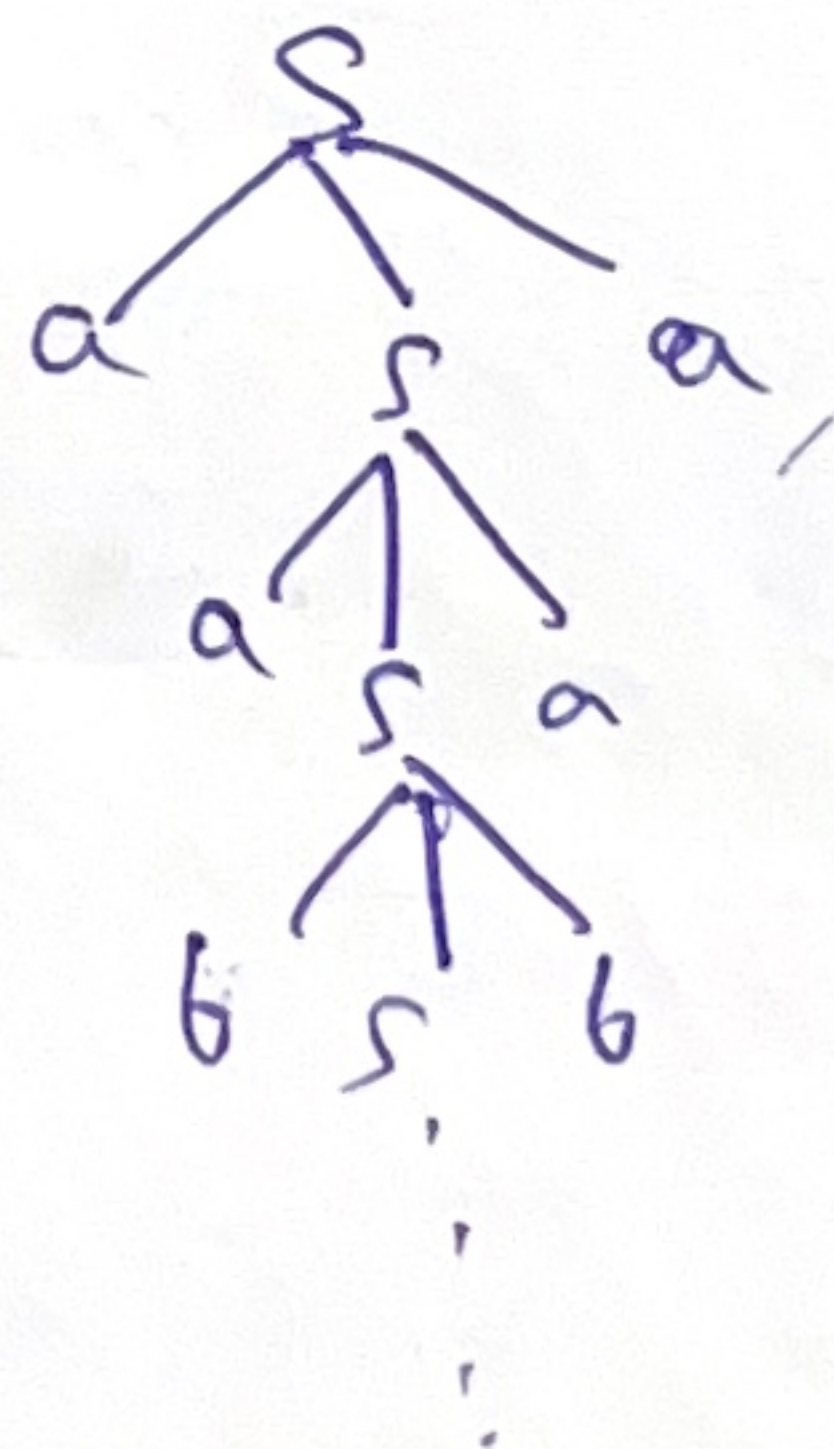
$$P_2: S \rightarrow bSb$$

$$P_3: S \rightarrow \epsilon$$

Palindrome

$$\underline{aabbbaa} \\ w \quad w^R$$

$$L = \{ww^R \mid w \in (a+b)^+\}$$



Prefix, Postfix, Palindrome, Grammar, Production Rule, Language, Derivation tree, Mathematical equation

if $w \in L(G)$, where G is CFG then w can be represented by a tree known as derivation tree or parse tree.

$S = aSbS \mid e \mid aa \mid bb$

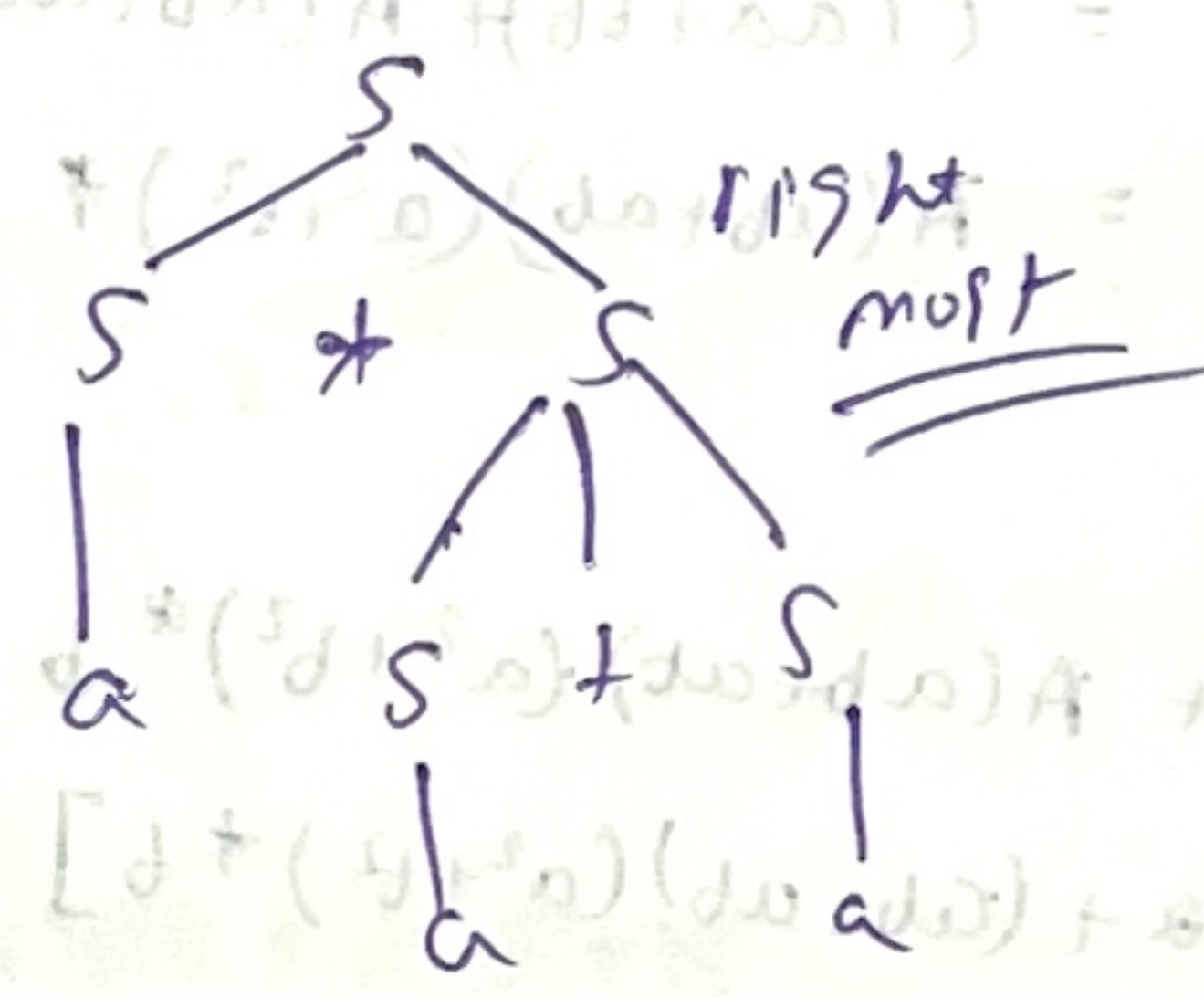
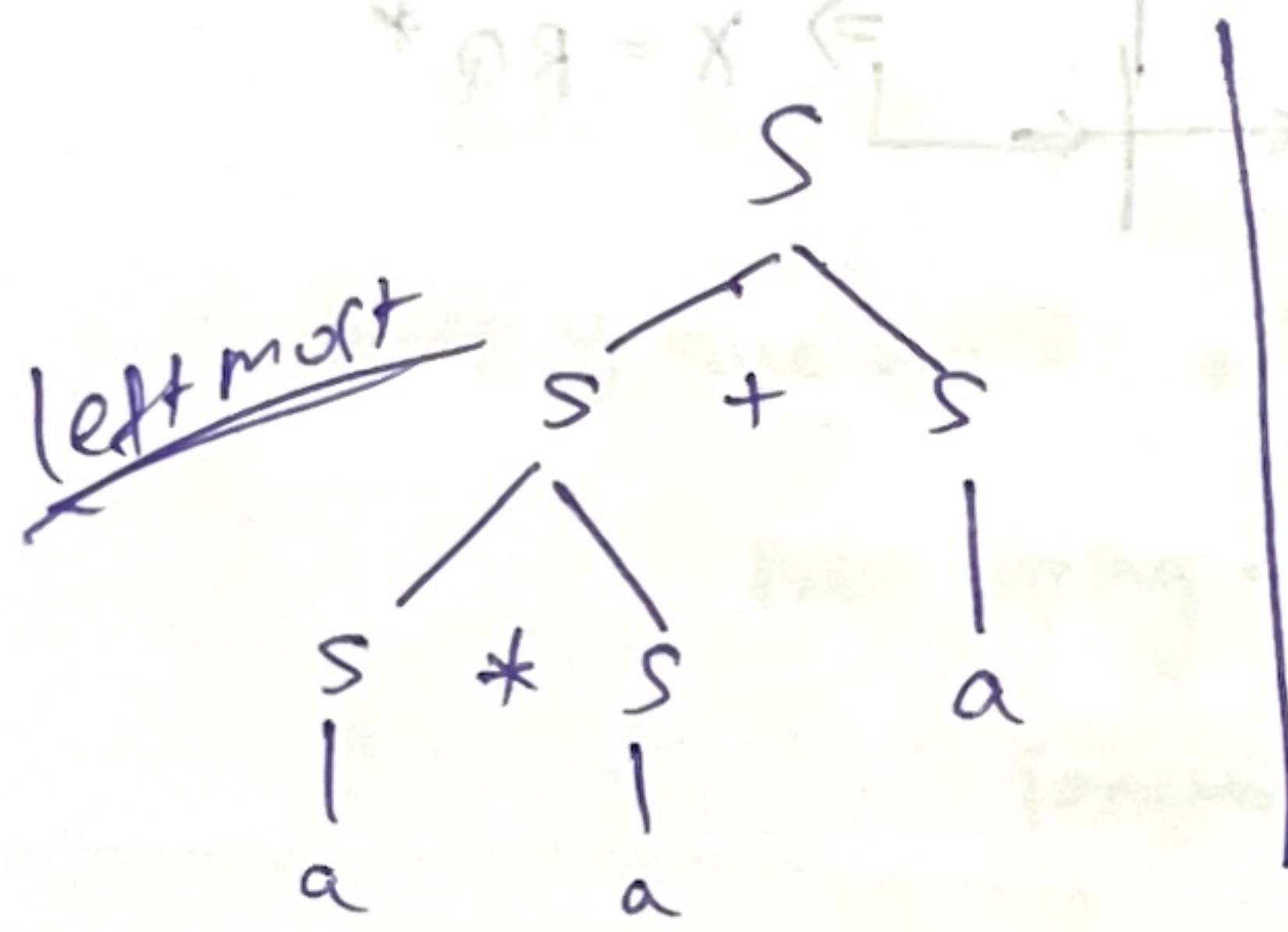
$w = aabbaabb$



1 to more
 option
 ↓
 Ambiguity

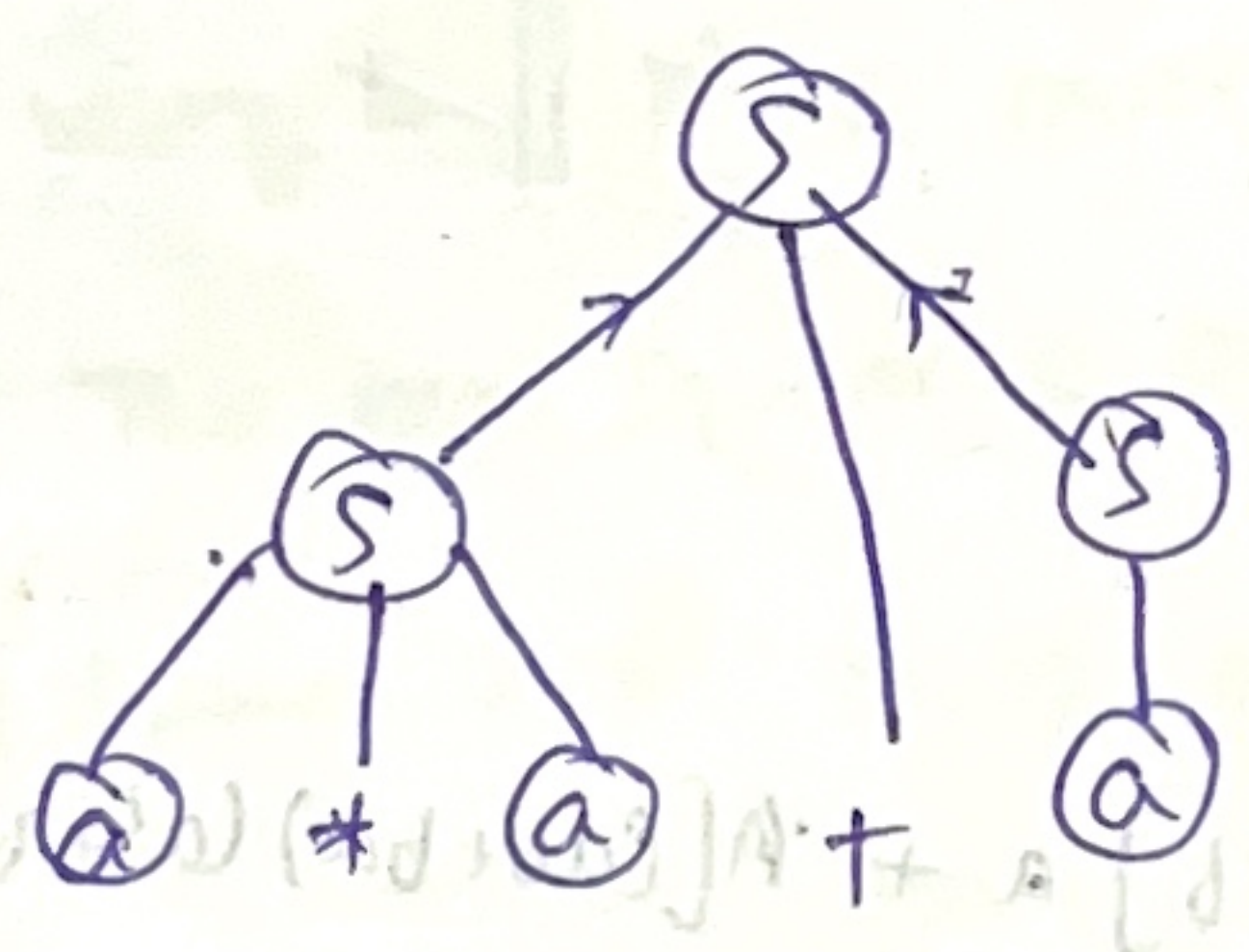
Q) $S \rightarrow S + S \mid S * S \mid a \mid b$

$w = a * a + a$



2 way derivation
 1) root $\rightarrow$ terminal
 2) terminal $\rightarrow$ root

2nd way



$w = aaaa bbb aaaa$

$S \rightarrow ASA \mid BSB \mid a \mid b$



Derivation Approaches

- 1) Start with starting sentential form continue this process until all the variable are replaced by terminal.

example

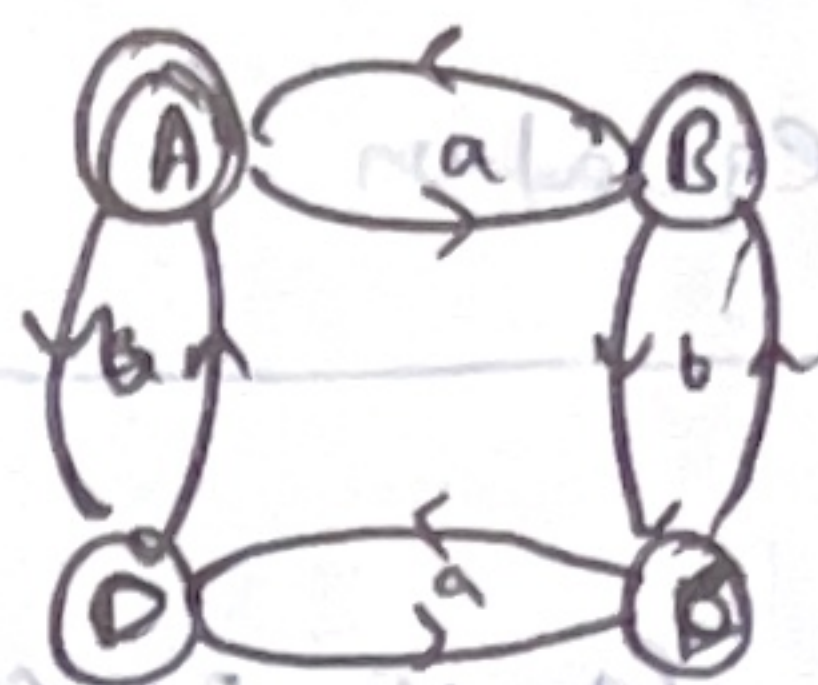
- 2) Recursive inference approach.

example.

Take w and check if $w \in L$ if the inverse ending (S) then $w \in L$

1) State

	a	b
A	B	D
B	A	C
C	D	B
D	C	A



$$A = Ba + Db + \epsilon$$

$$B = Aa + Cb$$

$$C = Bb + Da$$

$$D = Ca + Ab$$

$$\begin{aligned} C = Bb + Da &= (Aa + Cb)b + (Ca + Ab)a \\ &= Aab + Cbb + Caa + Aab \\ &= C(aa + bb) + A(ab + ab) \end{aligned}$$

$$C = A(ab + ab)(a^2 + b^2)^*$$

$$B = Aa + Cb$$

$$= Aa + A(ab + ab)(a^2 + b^2)^* b$$

$$= A[a + (ab + ab)(a^2 + b^2)^* b]$$

$$D = Ca + Ab$$

$$= A(ab + ba)(a^2 + b^2)^* a + Ab$$

$$= A[(ab + ba)(a^2 + b^2)^* a + b]$$

$$A = \epsilon + Ba + Db$$

$$= \epsilon + A[a + (ab + ba)(a^2 + b^2)^* b]a + A[(ab + ba)(a^2 + b^2)^* a + b]b$$

$$= \epsilon + A[a^2 + b^2 + (ab + ba)(a^2 + b^2)^* (ba + ab)]$$

using Arden's:

$$A = [a^2 + b^2 + (ab + ba)(a^2 + b^2)^* (ba + ab)]^*$$

This is the regular expression for DFA.

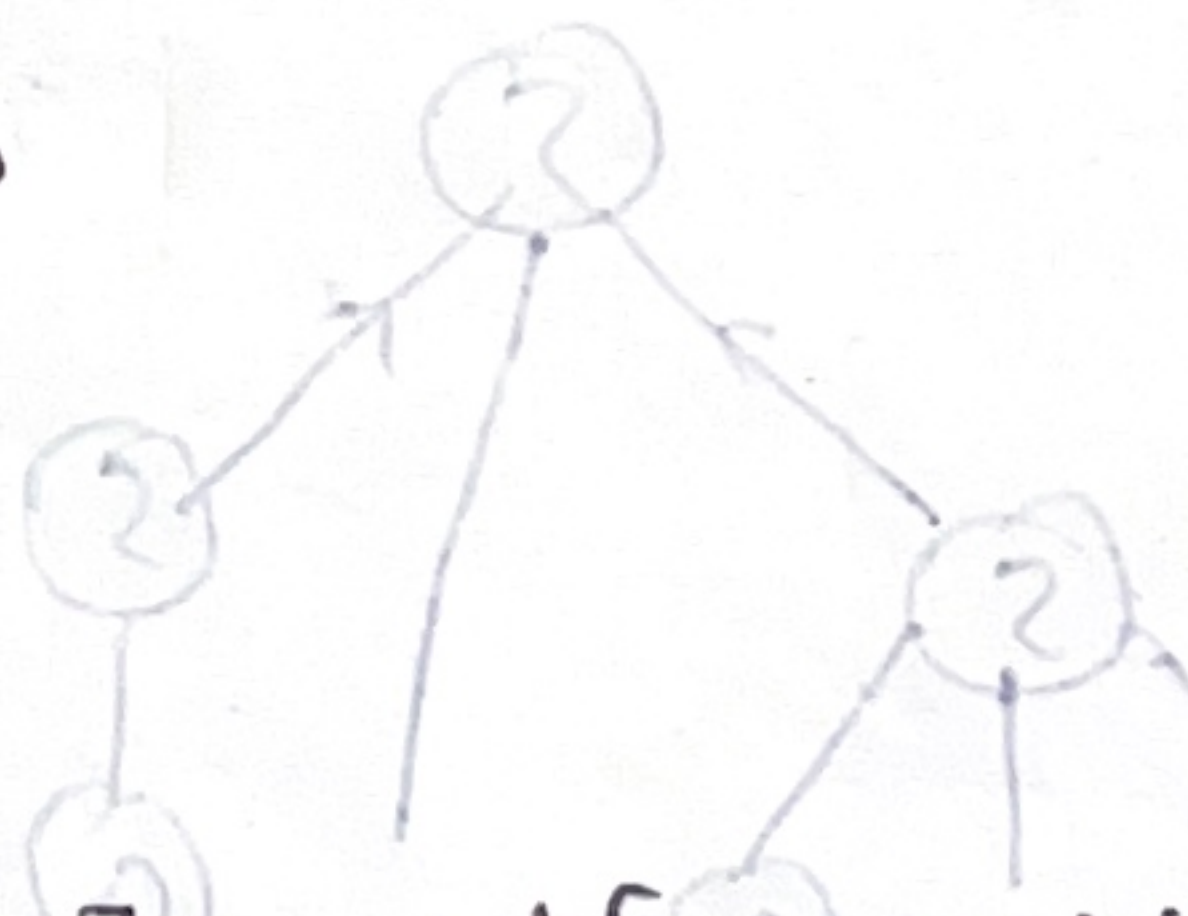
$$d^2 / a^2 / b^2 / c^2 / d^2 = 2$$

$$d^2 a^2 b^2 c^2 d^2 = 2$$

using Arden's theorem

$$X = XQ + R$$

$$\Rightarrow X = RQ^*$$



Q2) $L = \{a^i : i \geq 1\}$ is not RE

Let L is regular

then there exist a pumping length P

$$w = a^P$$

Since $P^2 > P$ the Pumping lemma applies

$$w = xyz \quad \text{where } |xy| \leq P$$

$$|y| \geq 1$$

$$w = a^P \quad \forall i \geq 0 \quad xy^iz \in L$$

first P characters are all a .

$$y = a^k \quad \text{where:}$$

$$1 \leq k \leq P$$

Pump y once more, Take: $i=2$

New string xy^2z

$$\text{length} \Rightarrow |xy^2z| = P^2 + k$$

for string to belong L $P^2 + k$ must be a perfect square.

$$\text{The next square is } (P+1)^2 = P^2 + 1 + 2P$$

$$\text{So any square } > P^2 \text{ must be } \geq P^2 + 2P + 1$$

$$\text{but } k \leq P$$

$$\text{So } P^2 + k < P^2 + 2P + 1$$

So $P^2 + k$ is not a perfect square

$$\text{So } xy^2z \notin L$$

this contradicts the pumping lemma condition.

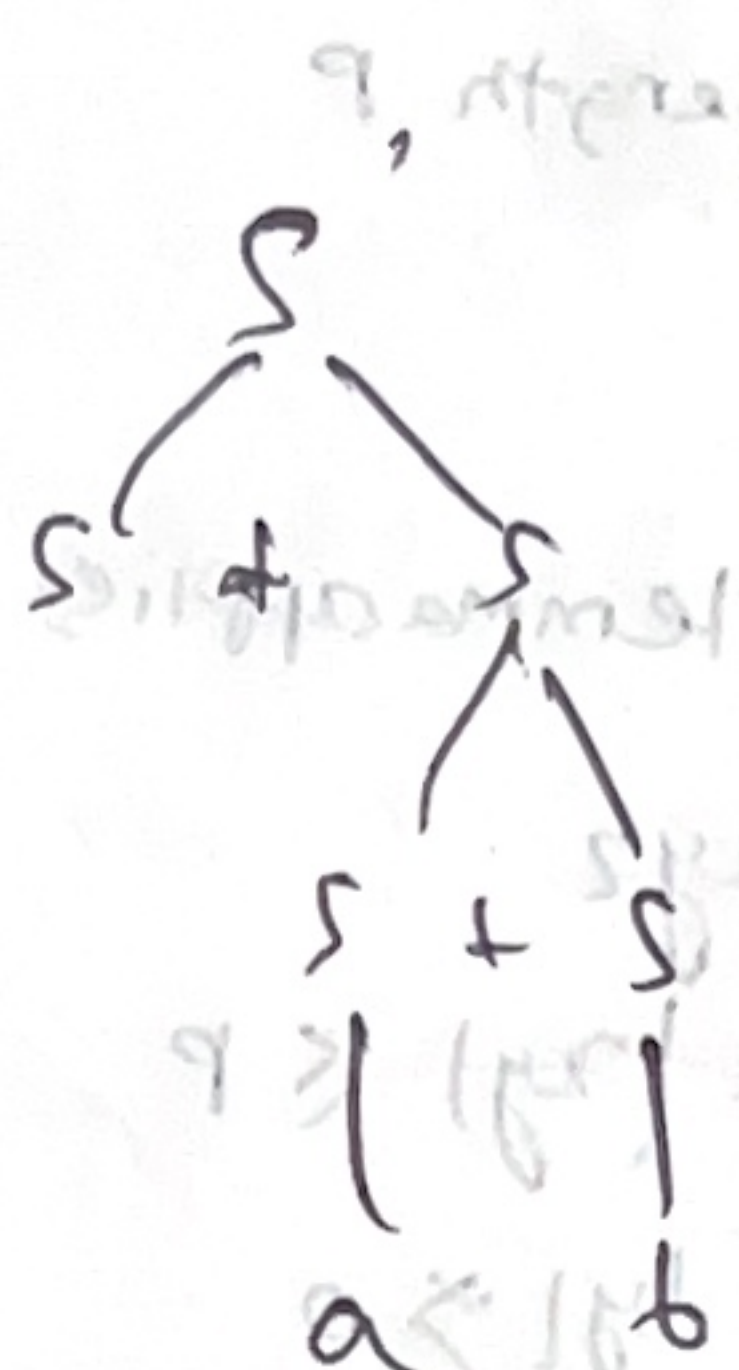
Hence our assumption is false

$$S = S + S \mid S * S \mid a \mid b$$

$$\text{and } w = a * a + b$$

⇒ left most

$$\begin{aligned} S &\Rightarrow S * S \\ &\Rightarrow a * S \\ &\Rightarrow a * S + S \\ &\Rightarrow a * a + S \\ &\Rightarrow a * a + b \end{aligned}$$



$$\begin{aligned} S &\Rightarrow S + S \\ S &\Rightarrow S * S \\ S &\Rightarrow a \\ S &\Rightarrow b \end{aligned}$$



Right most

$$\begin{aligned} S &\Rightarrow S + S \\ &\Rightarrow S + b \\ &\Rightarrow S * S + b \\ &\Rightarrow S * a + b \\ &\Rightarrow a * a + b \end{aligned}$$

Ambiguity

A given grammar (G) is ambiguous if there exist some string $w \in L(G)$ for which there are 2 or more derivation tree for a given string. (Parse tree)

Left Recursion

* in left recursion → chances of ∞ loop.

$$A \rightarrow A \alpha$$

$$A \rightarrow A \alpha_1 \mid A \alpha_2 \mid A \alpha_3 \mid \dots \mid B_1 \mid B_2 \mid B_3 \mid \dots$$

$$A \rightarrow \alpha_1 A' \mid \alpha_2 A' \mid \alpha_3 A' \mid B_1 \mid B_2 \mid \dots$$

$$A' \rightarrow B_1 A' \mid B_2 A' \mid B_3 A' \mid \epsilon$$

My ∞ loop
 $A \rightarrow A \alpha \alpha \alpha \dots A \alpha \alpha \alpha \dots A \alpha \alpha \alpha \dots$
 eventually stop
 $A \rightarrow B$
 or
 $A \rightarrow B \alpha$
 or
 $A \rightarrow B \alpha \alpha \dots$
 or
 $A \rightarrow B \alpha \alpha \dots$
 or
 $A \rightarrow B \alpha \alpha \alpha \dots$

$$\begin{aligned} \text{Ex: } A &\rightarrow b A' \\ \text{if } A' &= \epsilon \\ A &\rightarrow b \\ \text{if } A' &= + a A' \\ A &\rightarrow b + a A' \end{aligned}$$

$$A \rightarrow b, b + a, b + a + a, \dots$$

$$\begin{aligned} A &\rightarrow B A' \quad (A' \rightarrow \alpha \alpha \dots) \\ A' &\rightarrow \alpha A' \mid \epsilon \end{aligned}$$

$$E \rightarrow E + T / T \quad \begin{cases} E \rightarrow TE' \\ E' \rightarrow + TE' \end{cases}$$

$$T \rightarrow T * F / F$$

left factoring

$$A \rightarrow \alpha\beta_1 / \alpha\beta_2 / \alpha\beta_3 / \dots / r_1 / r_2 / r_3 / \dots$$

$\Downarrow$

$$A \rightarrow \alpha A' / r_1 / r_2 / r_3 / \dots$$

$$A' \rightarrow \beta_1 / \beta_2 / \beta_3 / \dots$$

$$S \rightarrow aSa / a$$

$$S \rightarrow aSa / a$$

$\Downarrow$

$$S \rightarrow aS'$$

$$S' \rightarrow Sa / a$$

Ambiguity in CFG

$$S \rightarrow S + S / S * S / a / b$$

$\Downarrow$

$$S \rightarrow S + T / T$$

$$T \rightarrow T * F / F$$

$$F \rightarrow a / b / (S)$$

$$w = a + a * b$$

Content free grammar

$$A \rightarrow \alpha \mid \alpha \in (V \cup T)^*$$

no restriction

with restriction

Chomsky Normal form

Greibach " (CNF)

CNF

$$\begin{cases} A \rightarrow \alpha \\ A \rightarrow BC \end{cases}$$

CNF

$$\begin{cases} A \rightarrow \alpha \\ A \rightarrow \alpha A \end{cases}$$

$$\alpha \in V^*$$

- The Production system of CFG is $A \rightarrow \alpha$ or $A \rightarrow (V \cup \Sigma)^*$ (no restriction on R.H.S.)
- we convert the grammar to (NF) to lower complexity of automation. → [easier to implement]
- NF put restrictions on design of production in (CFG). → Chomsky and Greibach

$$S \rightarrow SS$$

$$S \rightarrow a$$

$$S \rightarrow b$$

$$A \rightarrow A\alpha_1 / A\alpha_2 / \beta_1 / \beta_2$$

$$A \rightarrow \beta_1 A' / \beta_2 A'$$

$$A' \rightarrow \alpha_1 A' / \alpha_2 A' / \epsilon$$

Remove ambiguity from

Non term $\rightarrow$ term or $\rightarrow$ 2 term

Non term $\rightarrow$ αV^* $\downarrow$ non term

introduce a new terminal A'

$$A \rightarrow \beta$$

tree)

P.

$\alpha, A\alpha\alpha, \alpha\alpha\alpha$

by stop

$\rightarrow \beta$

$\rightarrow \beta\alpha$

$\rightarrow \beta\alpha\alpha$

or $\rightarrow \beta\alpha\alpha\alpha\alpha\alpha$

$(A' \rightarrow \alpha\alpha\alpha\alpha\alpha\alpha)$

$\alpha A' / \epsilon$

$\rightarrow 0$ times

$$S \rightarrow S^* a | b$$

these are the terminals

remaining part of the left recursion

$$S \rightarrow a S' | b S'$$

$$S' \rightarrow S S' | \epsilon$$

new terminal

Q) Remove the ambiguity.

$$G = \{S \rightarrow ASb | AS | \epsilon, A \rightarrow aA | \epsilon\}$$

common (A) (left)

$$S \rightarrow AS'$$

$$S' \rightarrow Sb | S$$

$$A \rightarrow aA | \epsilon$$

use formula 2

left factoring

or common factoring

True Ans

$$S \rightarrow ASb | AS | \epsilon, A \rightarrow aA | \epsilon$$

$$S \rightarrow AB | \epsilon, B \rightarrow Sb | S$$

new terminal

Again left factoring

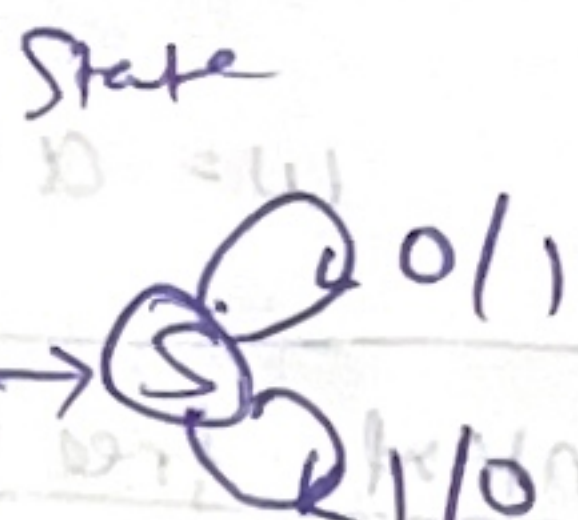
$$B \rightarrow SC, C \rightarrow b | \epsilon$$

$$A \rightarrow aA | \epsilon$$

Q) Consider a finite state machine which change

1's into 0 and 0's into 1.

input	output
0	1
1	0



State	0	1
N.st	S	S
O/P	1	0
Next State	S	S

Q) 2's complement of binary no. Reading from LSB (ex: 1101110)

011 011 110
100 100 111
011 100 101

first complement

add 1

1 0 0
0 1 1
+ 1
1 0 0

100
011

100
011
100

* Possible defects in CFG and their Removal

For the purpose of lang. generation or validation defect in grammar has to be removed. defect in grammar may cause following problems.

- i) It may restrict the grammar from generating string w.
- ii) It may contain 1 or more irrelevant syntactic category which may never be used.
- iii) it may involve illegal operations
- iv) it may involve multiple names and definition for same syntactic category

leading to unnecessary confusion.

① Ex- $T \rightarrow aaB | abA | aAT$ $w = "aabbba"$

$A \rightarrow aA$

$B \rightarrow ab | C$

$C \rightarrow ad$

Here variable 'c' is useless because it never occur in derivation stage. $C \rightarrow ad$ X

ii) $A \rightarrow aA$ } this is irrelevant (useless)

X $\rightarrow$ because it go in infinite loop.

$\rightarrow$ no termination condition [A symbol can be useless if it doesnot appear on the RHS of Production

Final Result $\Rightarrow B \rightarrow ab | C$

or
It donot take part in derivation of new String

* Elimination of null Production

$S \rightarrow aS | A$

$A \rightarrow \epsilon$

$S \rightarrow aS | A | a | \epsilon$

$L = \{a^n : n \geq 0\}$

from there 2 $S \rightarrow aS | \epsilon$

* Removal of unique Production

$S \rightarrow AB$, $A \rightarrow a$, $B \rightarrow c | b$, $c \rightarrow D$, $D \rightarrow E$, $E \rightarrow a$

we get $B \rightarrow a$

Final $= S \rightarrow AB$, $A \rightarrow a$, $B \rightarrow a | b$

Q) Remove unique Production

$S \rightarrow ABCD$

$A \rightarrow a$

$B \rightarrow \epsilon | b$

$C \rightarrow D$

$D \rightarrow \epsilon$

$S \rightarrow ABCD$

$A \rightarrow a$

$B \rightarrow \epsilon | b$

$C \rightarrow \epsilon$

cross = ϵ
or small ϵ

Q) Remove all null production such that $L(G_1) = L - \{\epsilon\}$

$S \rightarrow AAA|B$

$A \rightarrow OA|B$

$B \rightarrow \epsilon$

1) The only null production is $B \rightarrow \epsilon$

$\{w\} = \{B\}$

1) Remove $B \rightarrow \epsilon$

11) $w_2 = w_1 \cup x \mid x \in V$ and a variable in w_1 exist

in RHS production of x

so $S \rightarrow B$
and $A \rightarrow B$ } nullable

$\{B\} \cup \{A, S\} = \{A, B, S\}$

$A \rightarrow B, B \rightarrow \epsilon, A \rightarrow \epsilon$ | Remove
 $S \rightarrow B, B \rightarrow \epsilon, S \rightarrow \epsilon$ | Remove it

Final Production.

$S \rightarrow AAA|AA|A|C$

$A \rightarrow OA|O$

Cheers
the Ans
from Internet

1) CFL is super set of Regular language (CFL $\rightarrow$ Content free language) $CFL \supset RL$

11) if L_1 and L_2 are CFL then $(L_1 \cup L_2)$ is also CFL, $(L_1 \cdot L_2)$ is also CFL,

111) if L_1 is CFL $\Rightarrow (L_1^*)$ is also a CFL

1111) CFL are close over the operation string reversal

11111) CFL are not close over $(\cap)$, compliment, difference

$S \rightarrow BA$
 $A \rightarrow OAA$
 $B \rightarrow baBLb_1b_2|ba$
 $L = L_1^*$

It does not force to be in the form of A^*

not a CFL

not a CFL

$A \rightarrow BA$

$A \rightarrow C$

$A \rightarrow BA|C$

$L = \{0, 1\}^*$

not a CFL

$A \rightarrow B$

$A \rightarrow B, B \rightarrow C, C \rightarrow A$

$A \rightarrow B, B \rightarrow A$

$A \rightarrow BA$
 $A \rightarrow A$
 $B \rightarrow C$
 $C \rightarrow B$

$A \rightarrow BA$

$A \rightarrow C$

$A \rightarrow C$

Ex Convert a $G(\{S, A, B\}, \{a, b\}, P, S)$ with $P = \begin{cases} S \rightarrow aA, bB \\ A \rightarrow Ab, b \\ B \rightarrow Ba, b \end{cases}$

$$A \rightarrow (V \cup \epsilon)^*$$

~~A → a~~

to CNF

→ Prod system of CFG is in form of $A \rightarrow (V \cup \epsilon)^*$

→ NO restriction in RHS of Production

→ we convert G to NF to lower the complexity of the automation and easier to implement

→ A NF puts certain restriction on design of the production in the CFG

Two normal form

CNF (Chomsky)

GNF (Greibach)

Non terminal → Terminal

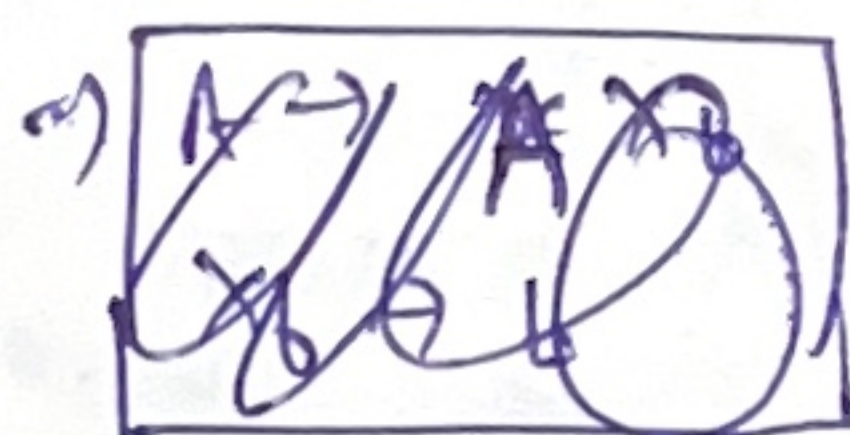
or
non terminal → 2 non terminal

non terminal → αV^*
↳ non terminal.

Q) $G(\{S, A, B\}, \{a, b\}, P, S)$

① Let Grammar $G \rightarrow$ Chomsky grammar

② i) $A \rightarrow Ab \rightarrow$ not in CNF



$$\begin{aligned} A &\rightarrow Ax \\ x &\rightarrow b \end{aligned}$$

Now 2 non terminal

→ rhs only terminal

→ CNF

ii) $B \rightarrow Ba$

Now $V_c = \{A, B, x_b\}$

$P_c \rightarrow \{A \rightarrow a, B \rightarrow a, x_b \rightarrow b, A \rightarrow Ax_b\}$

$$\begin{aligned} B &\rightarrow Bx_a \\ x_a &\rightarrow a \end{aligned}$$

$V_c = \{A, B, x_b, x_a\}$

$P_c \rightarrow \{A \rightarrow a, B \rightarrow a, x_b \rightarrow b, x_a \rightarrow a, A \rightarrow Ax_b, B \rightarrow Bx_a\}$

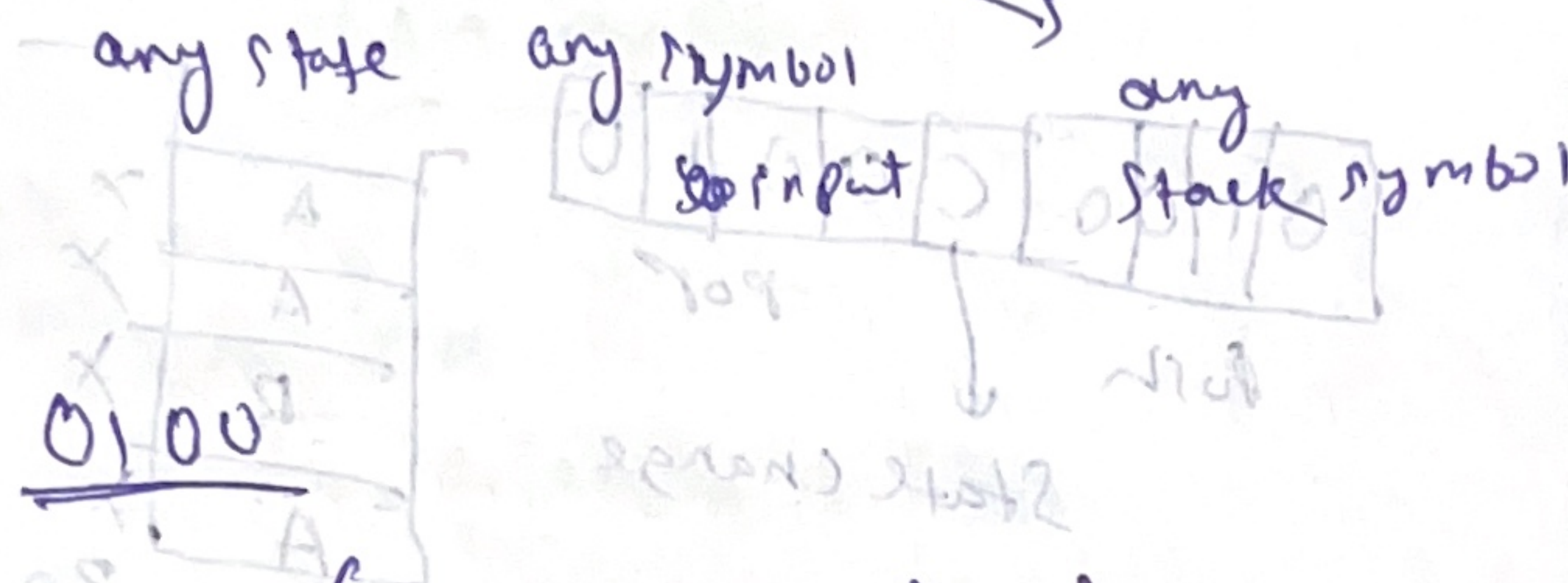
Epsilon

if the machine is in queue whenever in input take and headpoint the head will not move forward and change state

$$\sigma(q, \epsilon, A) = \{(p_1, r_1), (p_2, r_2)\}$$

$$\sigma: Q \times \Sigma \cup \{\epsilon\} \times \Gamma \rightarrow P(Q \times \Gamma)$$

X → Coors
Product

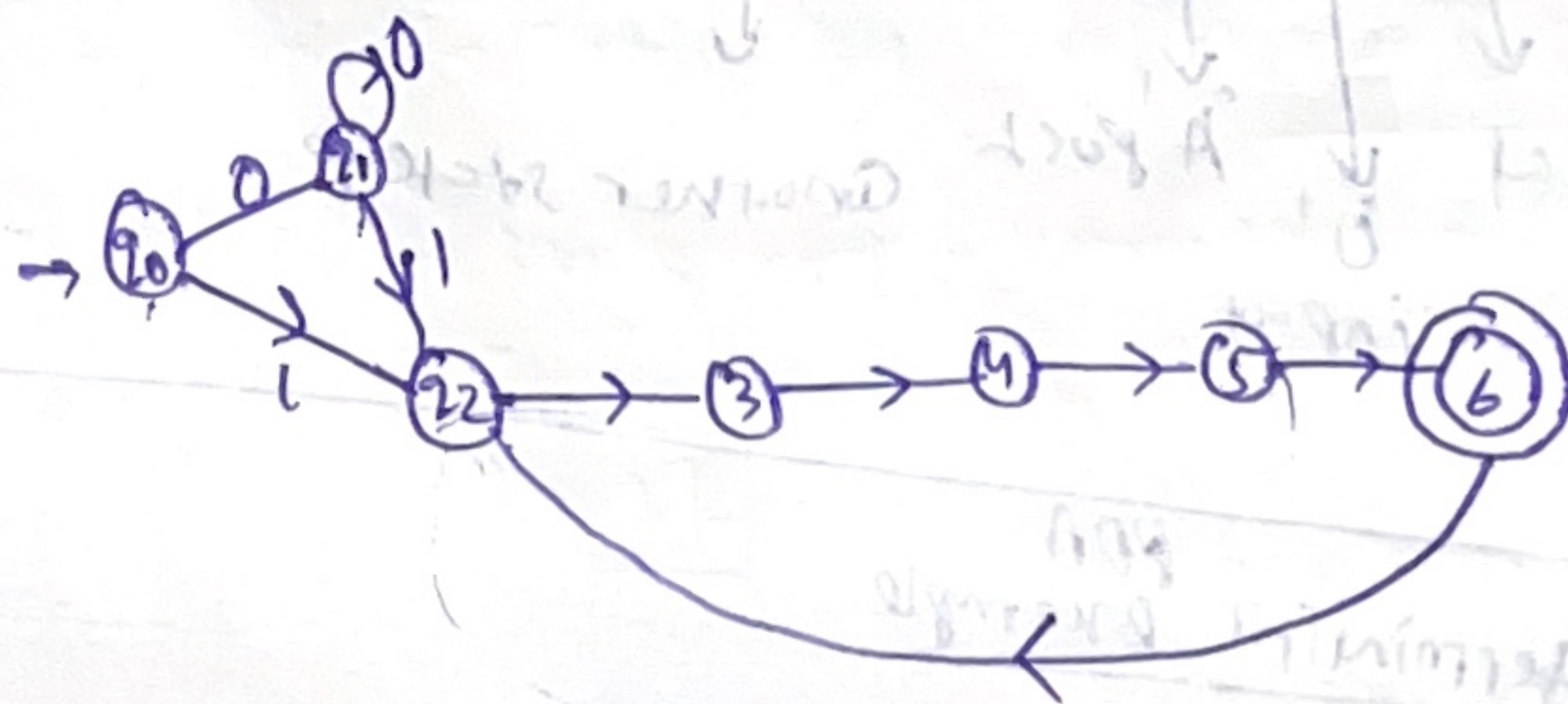
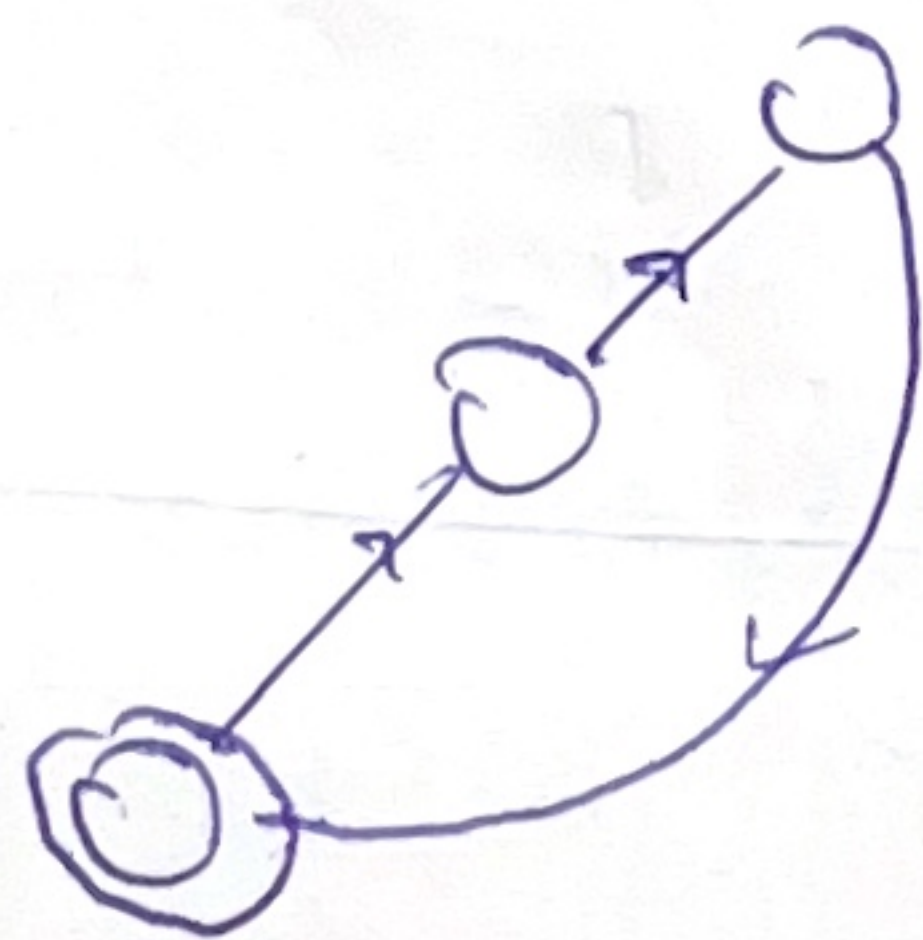
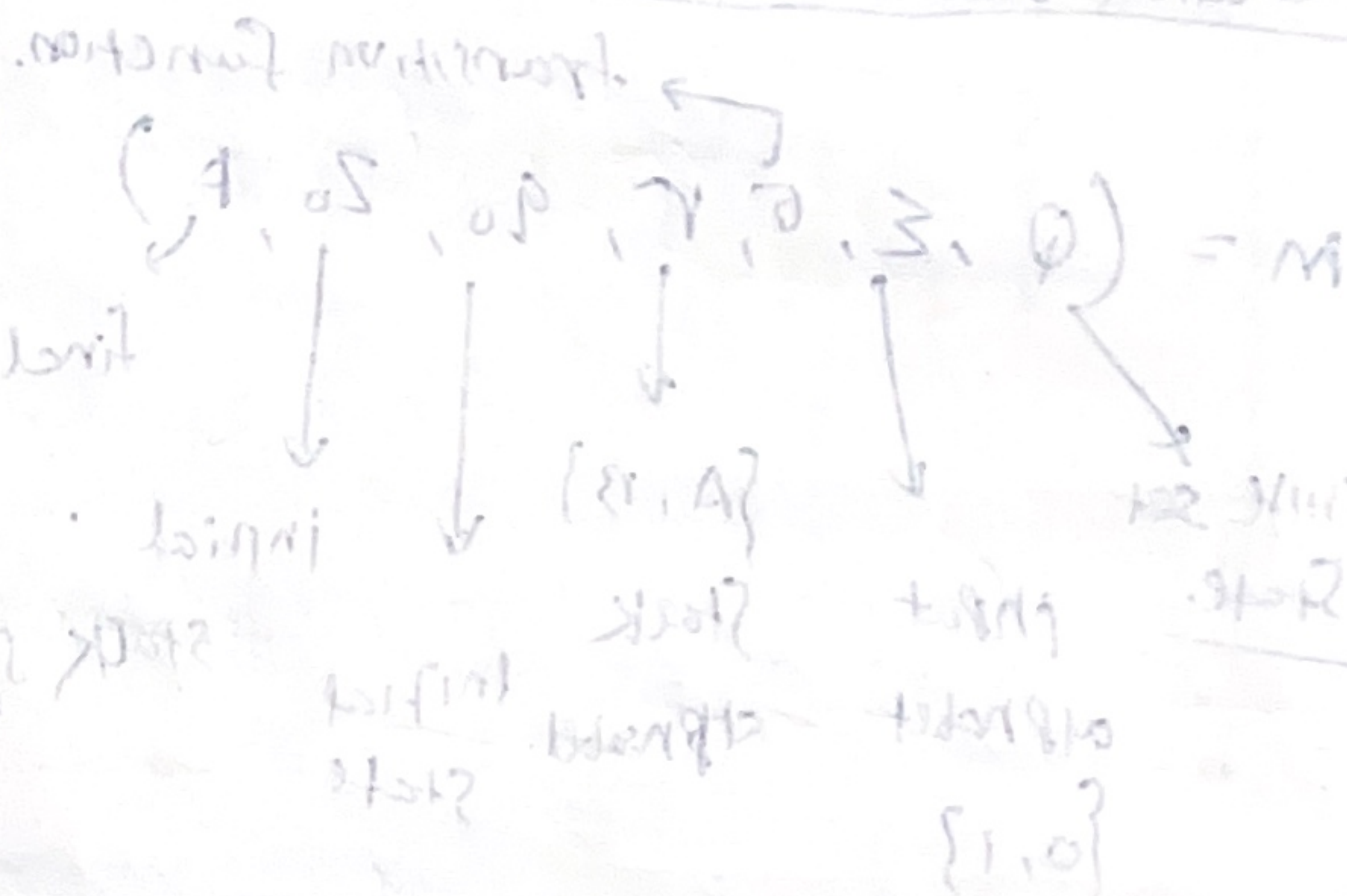


$$\sigma(q_0, 0, A) \rightarrow (0, A)$$

$$\sigma(0, 1, B) \rightarrow (1, B)$$

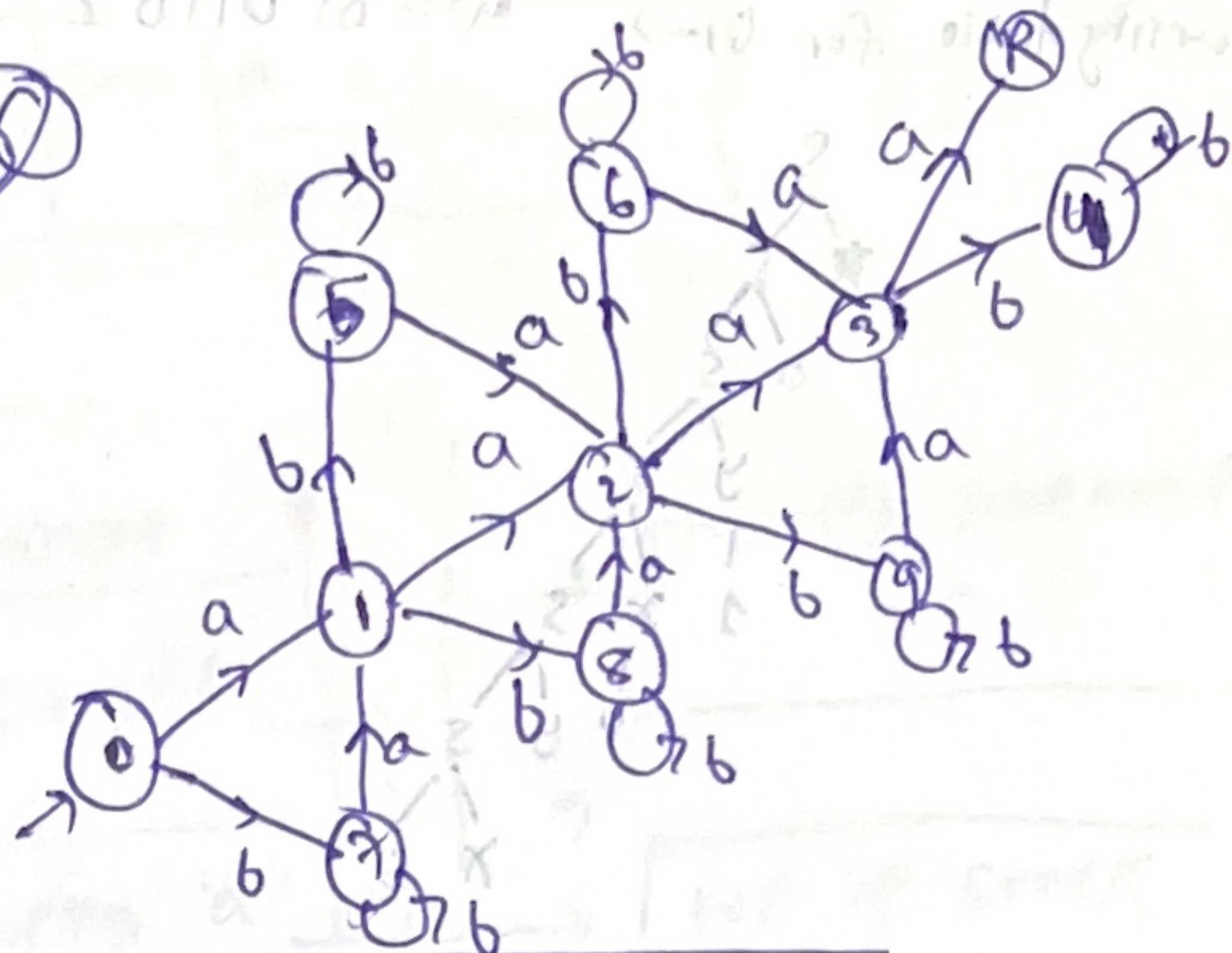
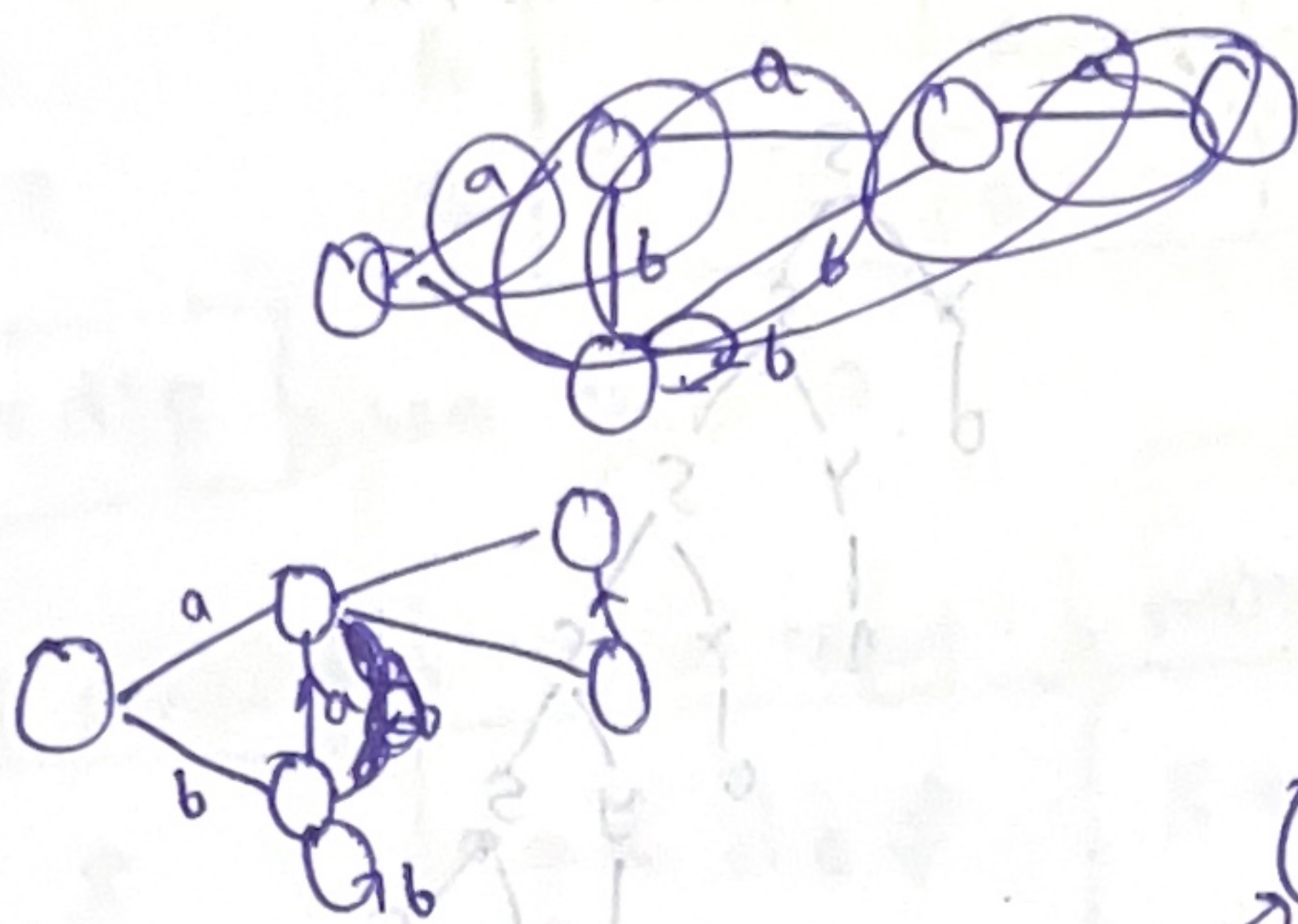
$$\sigma(0, 0, A) \rightarrow (0, A)$$

$$\sigma(0, 0, A) \rightarrow (0, A)$$



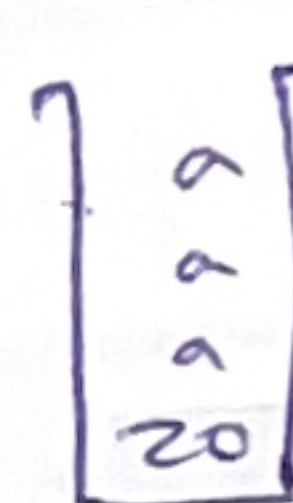
$$\sigma(q, \epsilon, A) = \{(p_1, r_1), (p_2, r_2)\}$$

No more than 3 a

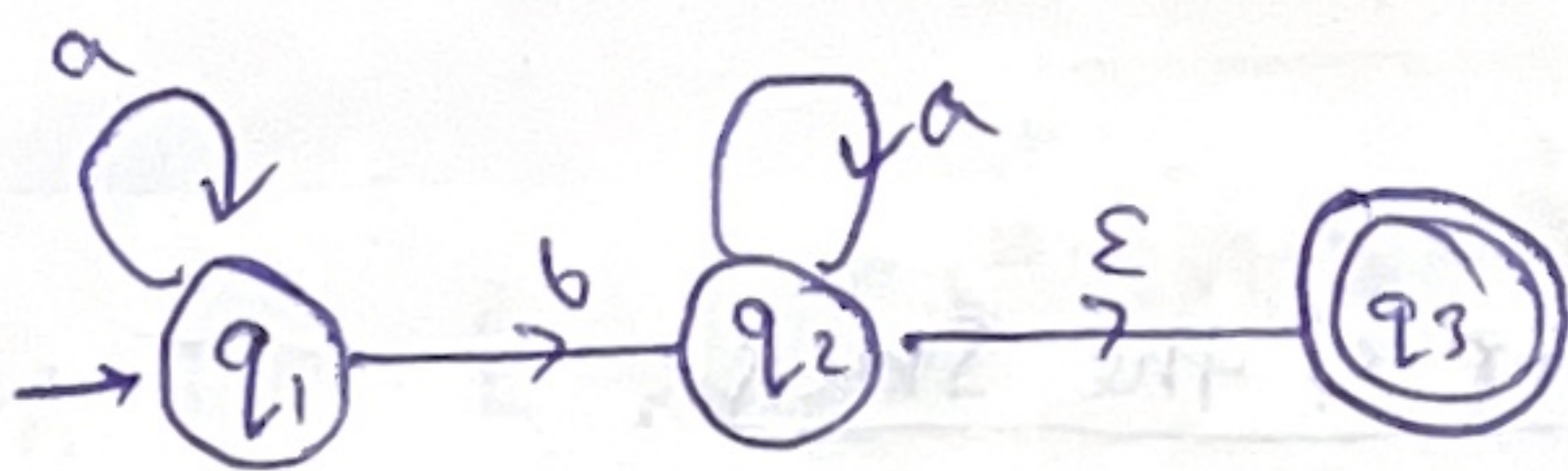


Design a PDA M_1 to accept the lang. $L_1 = \{a^n b a^n \mid n \geq 1\}$
 let $w = aaabaaa$

$\sigma(q_0, aaabaaa, z_0) = \sigma(q_1, aabaaa, az_0)$
 $\sigma(q_1, abaaa, aaz_0)$
 $\sigma(q_1, baaa, ciaaz_0) \rightarrow$
 $\sigma(q_2, aaa, aaz_0)$
 $\sigma(q_2, aa, aaz_0)$
 $\sigma(q_2, a, aaz_0)$
 $\sigma(q_2, \epsilon, z_0)$



Final state $\sigma(q_3, \epsilon, \epsilon)$



Parsing table
 and
 parsing Algorithm

Parsing
 Given a sentence belonging to a lang. the task of finding derivation is known

a) Parsing

for natural language brain perform the task of parsing to understand the underline meaning.

if a statement is not syntactically correct it can't parse by Compiler + Syntax error.

To check if $w \in L$, the PDA has to choose the appropriate transition function from the

Set of delta of the PDA. To automate the parsing process using the PDA there

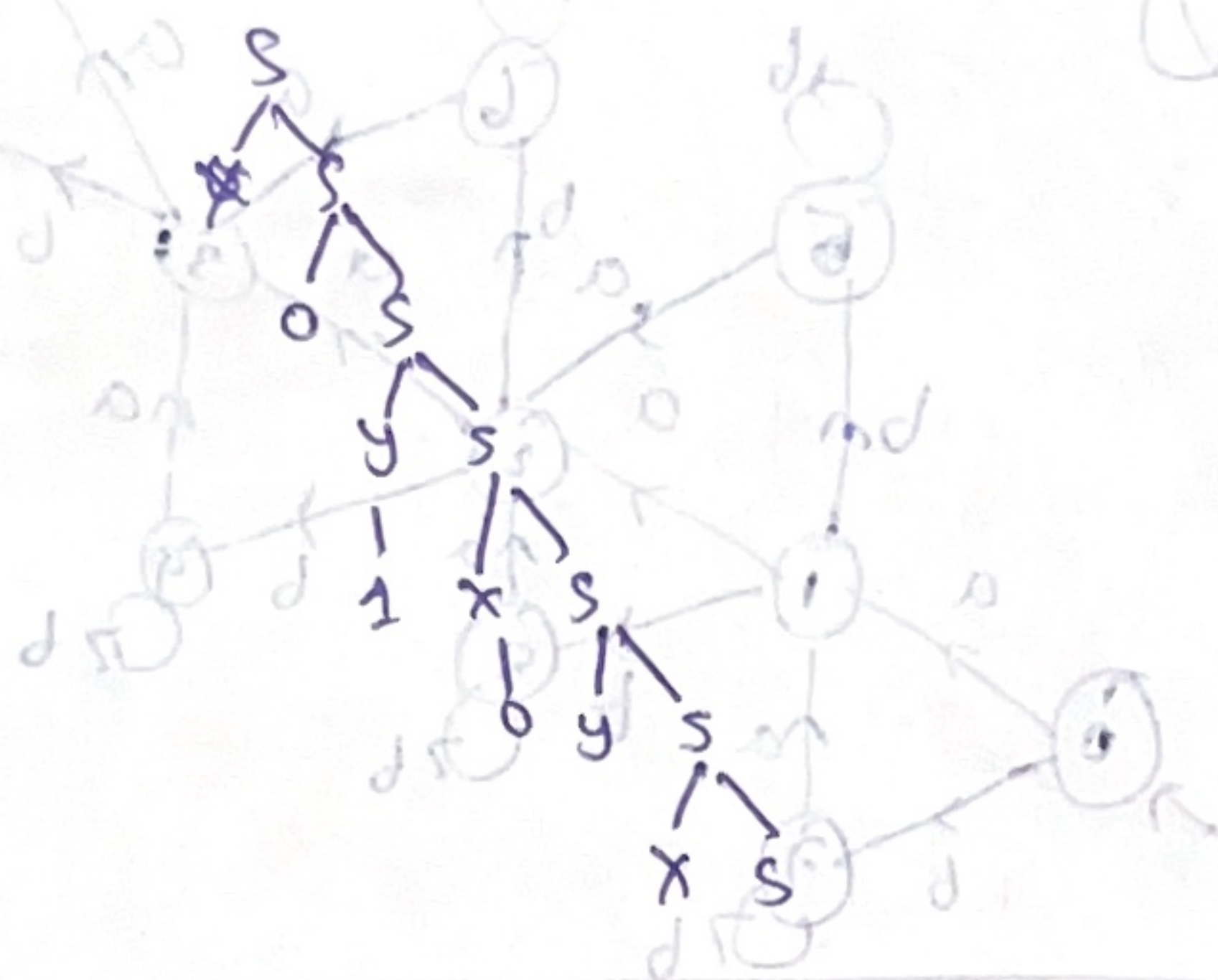
should be a ~~helping~~ guiding mechanism to let the PDA know which transition to

choose. The task is accomplished by creating guiding table $\rightarrow$ Parsing table.

$$X \rightarrow XS \mid YS \mid 2, X \rightarrow 0, Y \rightarrow 1$$

$$w = 0101102$$

Design
Parsing table for G1 →



$$P_1: S \rightarrow XS$$

$$P_2: S \rightarrow YS$$

$$P_3: S \rightarrow 2$$

$$P_4: X \rightarrow 0$$

$$P_5: Y \rightarrow 1$$

	ϵ	0	1	2
S	err	P1	P2	P3
X	err	P4	E	E
Y	err	err	P5	E

$$L = \{a^{2n} : n \geq 0\}$$

Context free language.

$$L = \{a^n b^n : n \geq 1, 2, 3, \dots, \infty\}$$

$$L = \{a^n b^n : n \geq 1\}$$

Parsing → top down (table-driven predictive parser)
bottom up

i) The Program consider 'X' is Symbol of top of the Stack,

next input symbol → 'a'

ii) if $X = a = \epsilon$ (then Parser announces successful completion of Parsing & h)

iii) if $X = a \neq \epsilon$ (then Parser Pop 'X' of the stack and advance the input pointer

iv) if 'X' = non terminal
Program consult Parsing table entry → $TABLE[X, A]$ → a Particular cell of table
Get the production entry
if its $X \rightarrow UVW$
the X replace by UVW
↓
top

if its error
Parser call error recovery
routine.

Q1)

$S \rightarrow aABb$

$A \rightarrow c|e$

$B \rightarrow d|e$

	a	b	c	d	\$
S	$S \rightarrow aABb$				
A		$A \rightarrow e$	$A \rightarrow c$	$A \rightarrow e$	
B		$B \rightarrow e$		$B \rightarrow d$	

empty are error

$w = acdb\$$ given check it

Stack Content	user input	moves
	$a c d b \$$	$S \rightarrow aABb$
$\$$	$a c d b \$$	Pushing 'a'
$\$ b B A$	$a c d b \$$	$A \rightarrow c$
$\$ b B A$	$c d b \$$	Pushing c
$\$ b B c$	$c d b \$$	$B \rightarrow d$
$\$ b B$	$d b \$$	Pop d
$\$ b d$	$d b \$$	Pop b
$\$ b$	$b \$$	successful completion of parsing
$\$$	$\$$	

Starting sentential = S

top of stack = matching

$\neq \$$ so pop

A (top of stack)

Symbol read = c

So $table(AC) = A \rightarrow c$

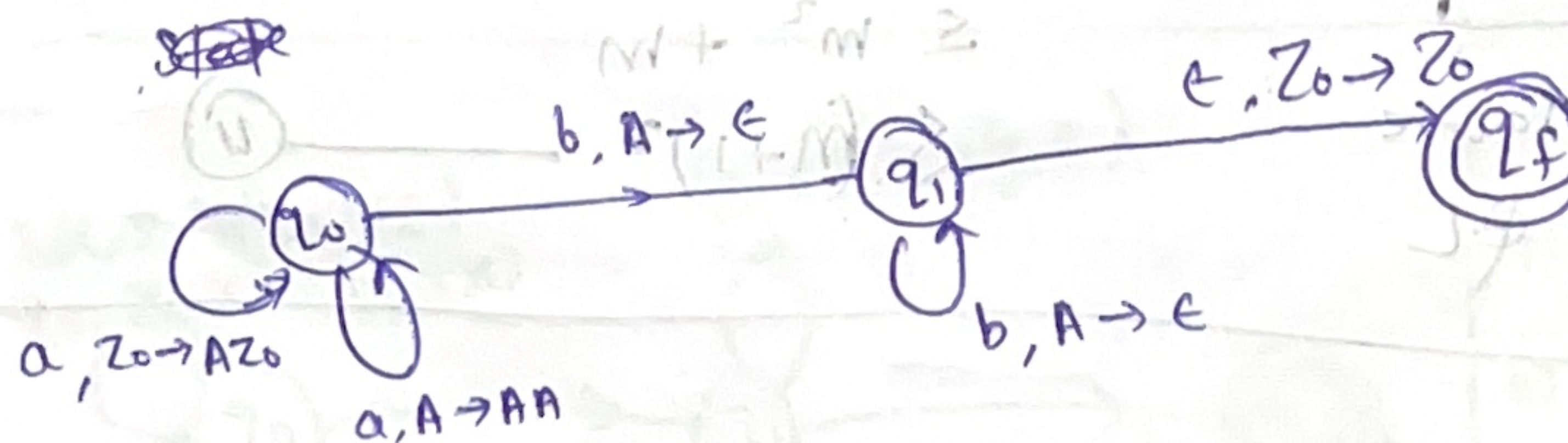
A Pop

C Push

$C = c \neq \$$

Pop

Q2) PPA for $L = \{a^n b^n : n \geq 0\}$



Ex: $\vdash (q_0, a, Z_0) \vdash (q_0, AZ_0)$

inp	State	top	operation	next
a	q_0	Z_0	$Z_0 \rightarrow AZ_0$	q_0
a	q_0	A	$A \rightarrow AA$	q_0
b	q_0	A	$A \rightarrow \epsilon$	q_1
b	q_1	A	$A \rightarrow \epsilon$	q_1
ϵ	q_1	Z_0	$Z_0 \rightarrow Z_0$	q_f

Q) $\{z = a^p \mid p \text{ is prime}\}$ is not CFL. $|x| \geq 1$; $|vwx| \leq n$

Let $L \subseteq CFL$ if p is prime such that $p > n$. $z = a^p \in L$

Let $z = uvwxy$ and $|z| = p$

Since $z = uvwxy \in L$, by Pumping Lemma

$uv^kwx^ky \in L$

$$\begin{aligned} |uv^kwx^ky| &= |uvwx| + (k-1)|vx| \\ &= p + (k-1)r \\ &= p + pr \end{aligned}$$

Let $|vx| = r$
Let $k = p+1$

$uv^kwx^ky \notin L$

$$= p(1+r) \rightarrow \text{composite no}$$

Q) $L = \{d \mid d \text{ is perfect square}\}$ is not CFL

Let L is CFL; $\exists$ constant n ; $\forall$ string $z \in L$ with $|z| > n$

$uvwx \in L$ and $|vwx| \leq n$ and $|vx| \geq 1$

for all $k \geq 0$; $k \in \mathbb{N}$, $uv^kwx^ky \in L$

Let m is the m^2 .

Not be perfect square, $z = 0^n \in L$

$z = uvwx$
 $uv^2wx^2y \in L$

$uv^2wx^2y \in L$

$|uv^2wx^2y| > |uvwx|$

$|uv^2wx^2y| > m^2$ ——— (1)

from (1) and (2)

$$m^2 \leq |uv^2wx^2y| \leq (m+1)^2$$

so it's not perfect square
is not CFL, $\notin L$

$$|uv^2wx^2y| = |uvwx| + |vx|$$

$$\leq |uvwx| + |vwx|$$

$$\leq m^2 + m$$

$$\leq (m+1)^2$$
 ——— (2)

for contradiction

if our contradiction is wrong

our opposite is true

a) $\{a^n b^n \mid n \geq 0\}$ is not regular

$$w = a^p b^p$$

$$w \in L \text{ and } |w| \geq p$$

$$w = xyz$$

$$|xy| \leq p$$

$$|y| > 0$$

$$xyz \in L$$

for every $i \geq 0$

$y = a^k \mid k > 0$
now pump y twice

$$xy^2z = a^{p+k} b^p$$

$$k > 0$$

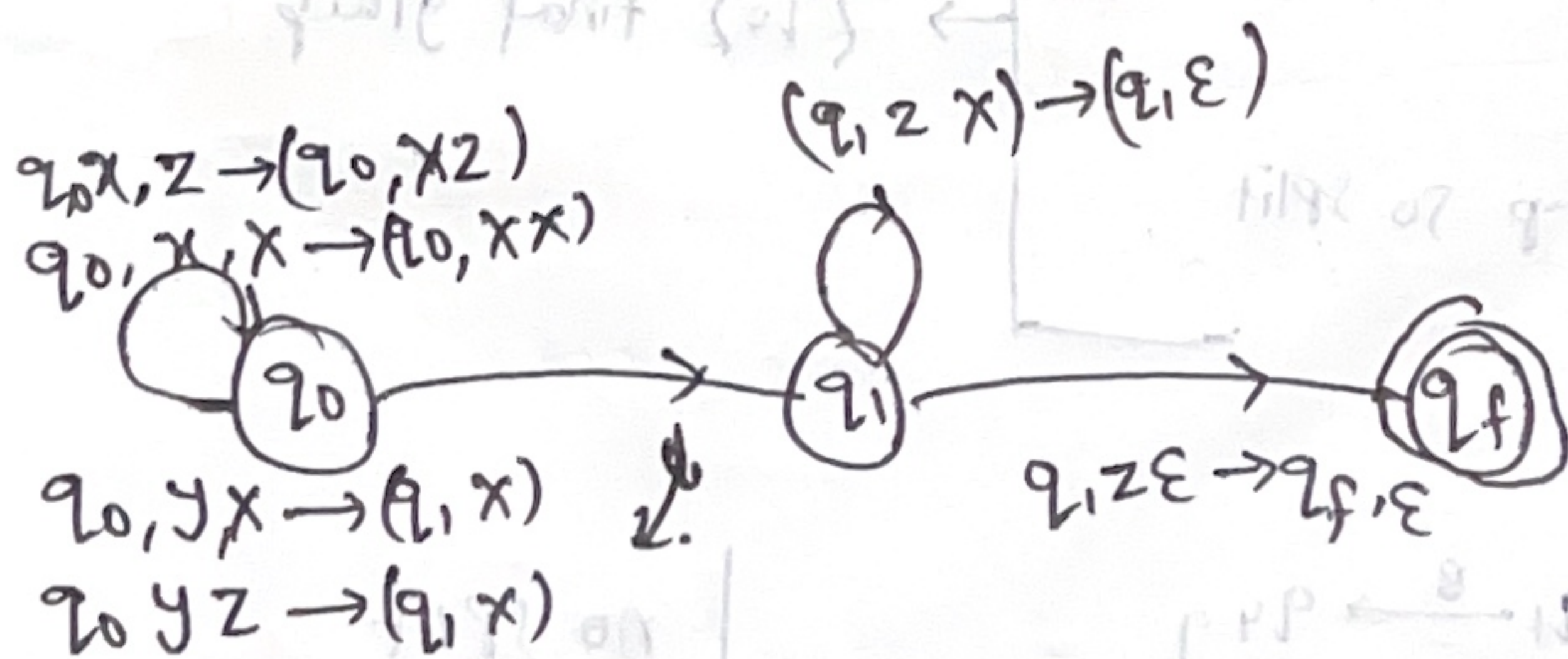
$$\text{so } p+k \neq p$$

$$\text{so } a^{p+k} b^p \notin L$$

so not Regular

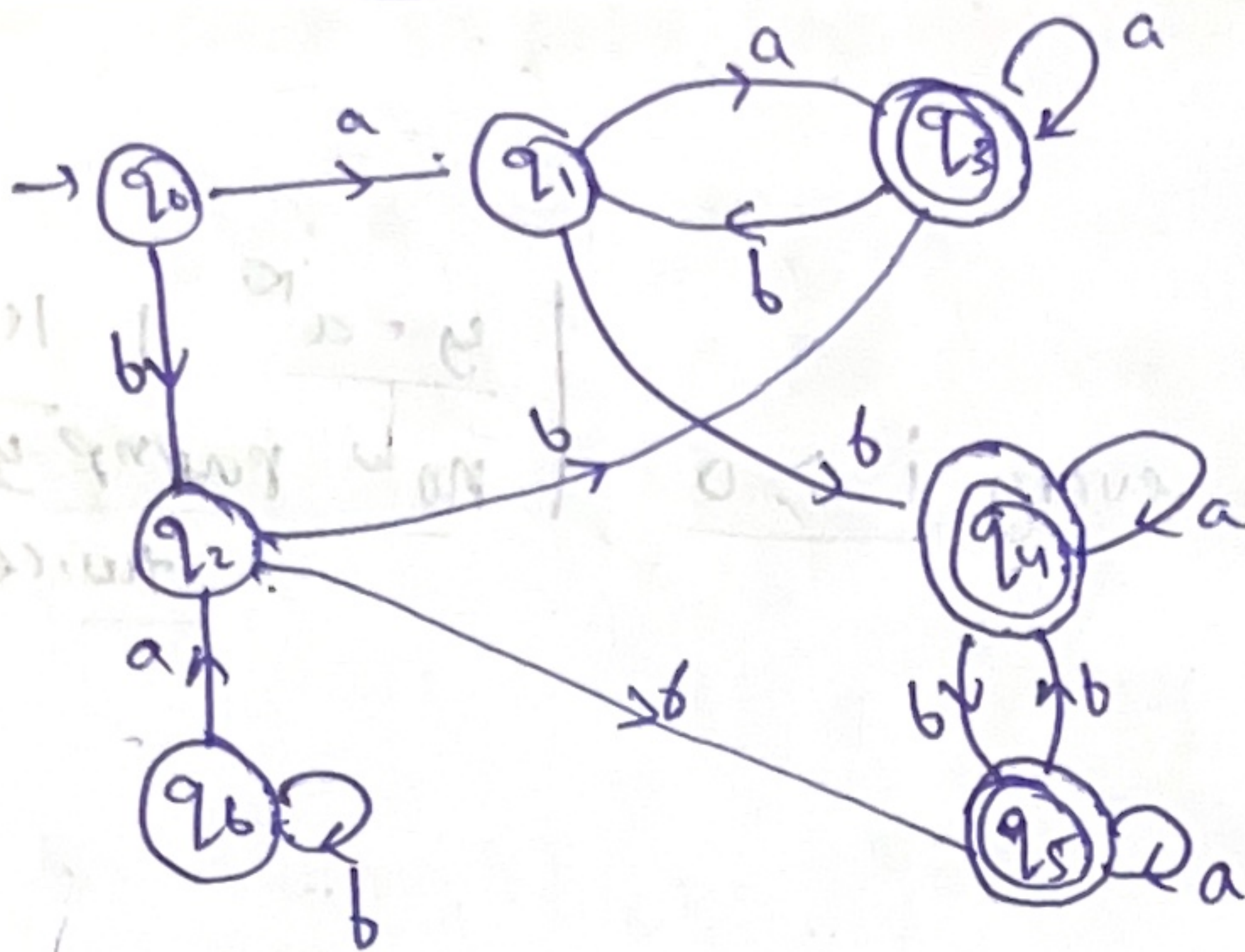
Q) PPA $\{x^n y z^n \mid n \geq 0\}$

Current State	Stack top	input	Stack operation	next state
q_0	z	x	push x	q_0
q_0	x	x	push x	q_0
q_0	x	y	do nothing	q_1
q_0	z	y	do nothing	q_1
q_1	x	z	pop x	q_1
q_1	z	ϵ	do nothing	q_f



- i) $(q_0, z, z) \rightarrow (q_0, x)$
- ii) $x (q_0, x, x) \rightarrow (q_0, xx)$
- iii) $y (q_0, y, x) \rightarrow (q_1, x)$
- iv) $z (q_1, z, x) \rightarrow (q_1, \epsilon)$
- v) $z (q_1, z, x) \rightarrow (q_1, \epsilon)$
- vi) $\epsilon (q_1, \epsilon, z) \rightarrow (q_f, \epsilon)$

Minimization of DFA



	a	b
q0	q1	q2
q1	q3	q4
q2	q3	q5
q3	q3	q1
q4	q4	q5
q5	q5	q1
q6	q2	q6

① Unreachable state $\rightarrow$ q_6 Remove it.

② equivalent

let assume all states are equivalent $\{q_0, q_1, q_2, q_3, q_4, q_5\}$

divide into $\rightarrow$ Accepting state $\{q_3, q_4, q_5\}$ # final state

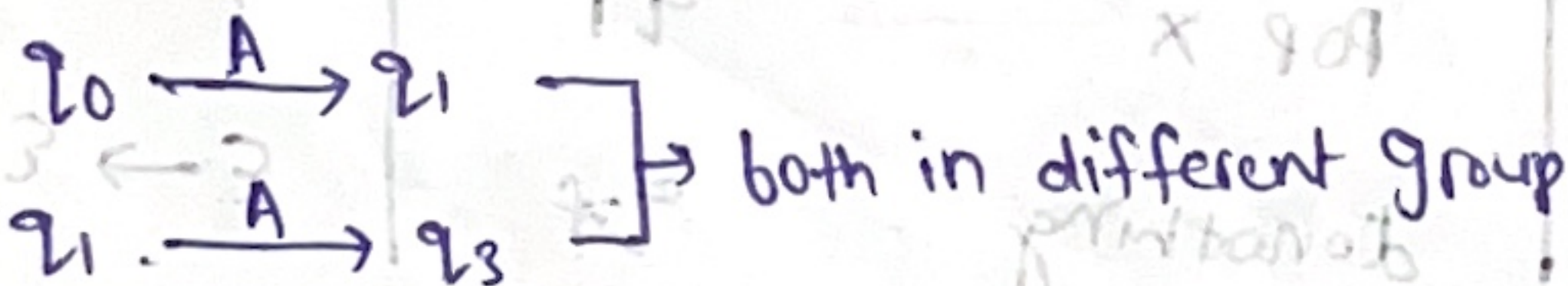
$\rightarrow$ non accepting state $\{q_0, q_1, q_2\}$ # non final state

representation

$$\Pi_0 = \{ \{q_0, q_1, q_2\}, \{q_3, q_4, q_5\} \}$$

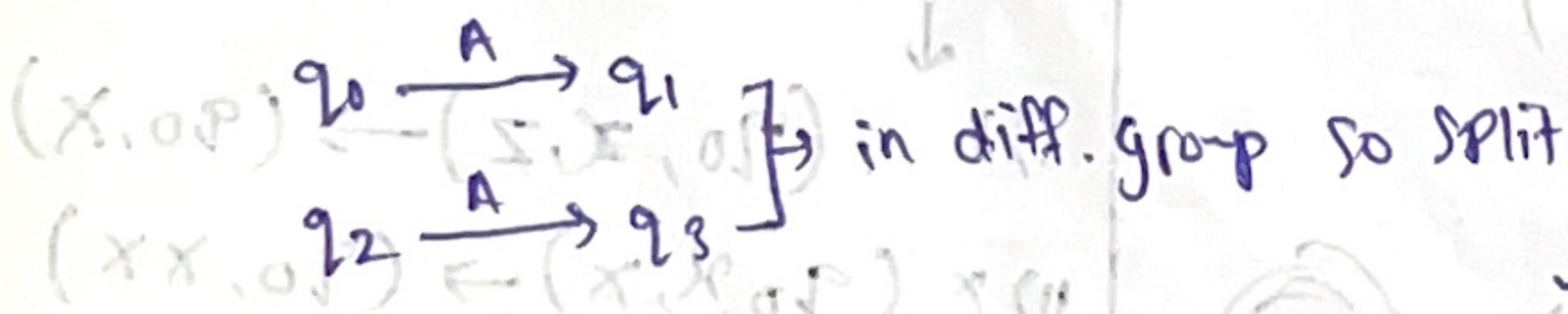
Note The transition from the state and if a transition enter into the same equivalent Partition State set, then we say that these sets of state containing equivalent state.

* compare q_0 and q_1

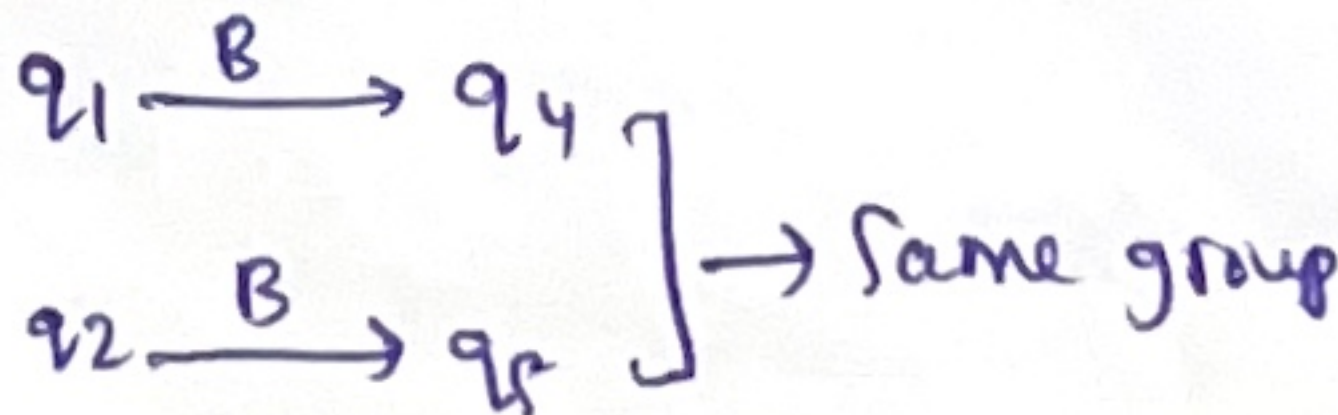
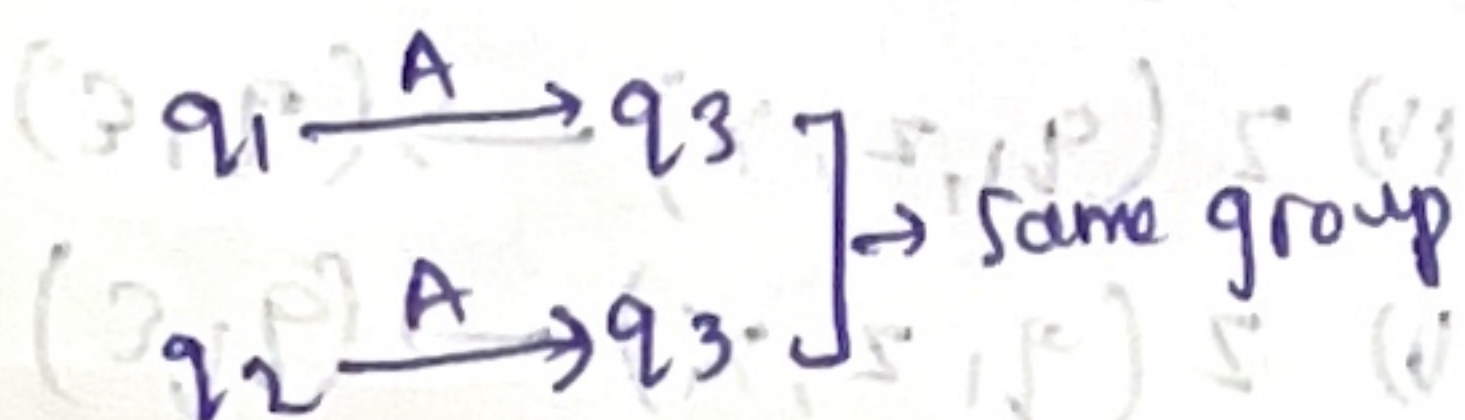


so they need to split

* Compare q_0 and q_2



* Compare q_1 and q_2



no split

$\{q_1, q_2\}$ final group

$\{q_0\}$ final group

$$\{G_1\} \rightarrow \{\{q_0\}, \{q_1, q_2\}\}$$

Similarly doing these steps for G_2

$$G_2 = \{\{q_3\}, \{q_4, q_5\}\}$$

$$G_2 = \{\{q_3\}, \{q_4, q_5\}\}$$

$$\Pi_1 = \{\{q_0\}, \{q_1, q_2\}, \{q_3\}, \{q_4, q_5\}\}$$

find Π_2 → can check for q_1, q_2

no split

final $\{q_1, q_2\}$

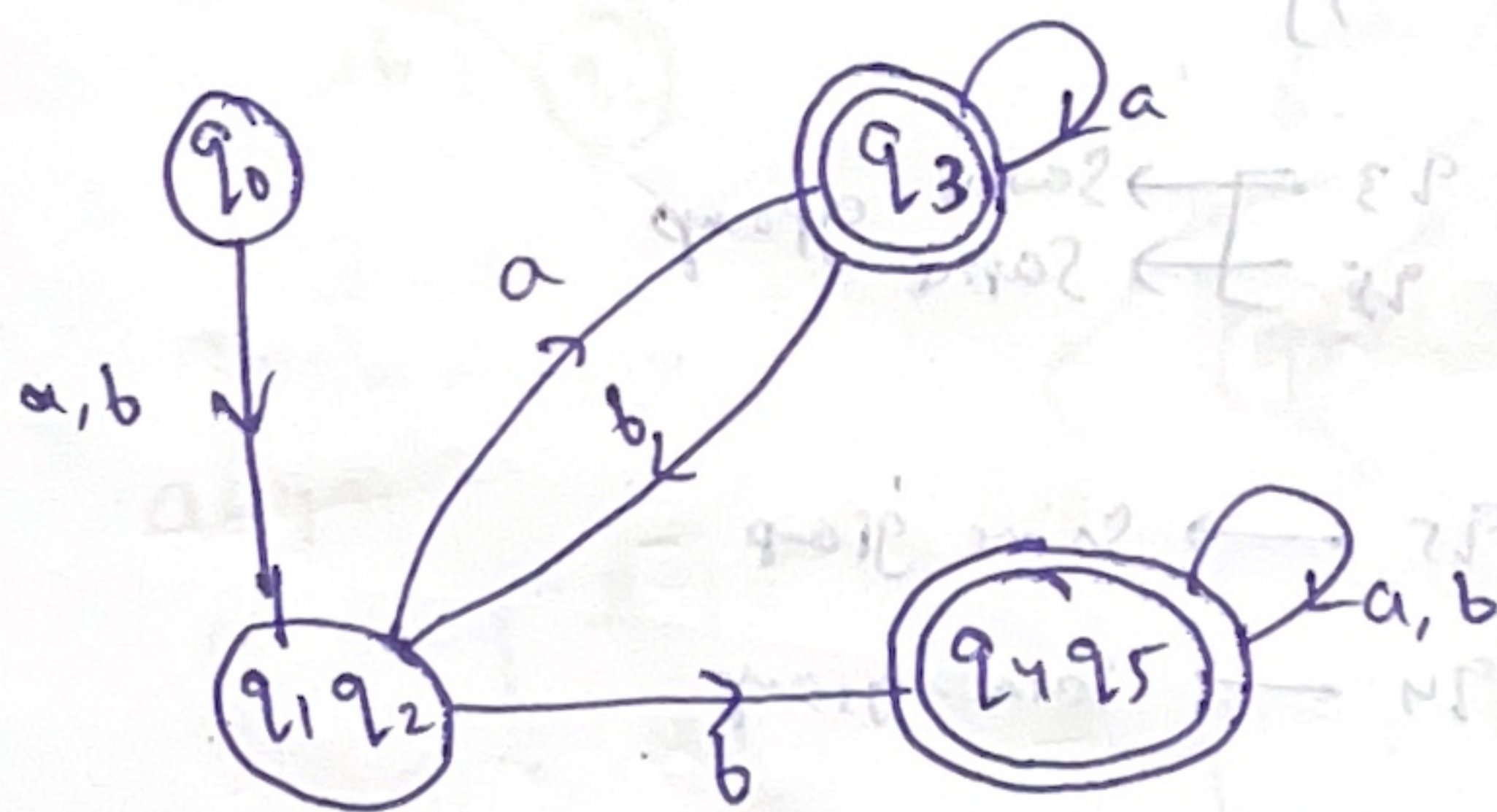
* check q_4, q_5

$$\begin{array}{c|c} q_4 \xrightarrow{A} q_4 & q_4 \xrightarrow{B} q_5 \\ q_5 \xrightarrow{A} q_5 & q_5 \xrightarrow{B} q_4 \end{array} \quad \begin{array}{l} \text{no split} \\ \text{because same group} \end{array} \quad \{q_4, q_5\} \text{ final}$$

$$\begin{array}{c} q_1 \xrightarrow{A} q_3 \\ q_2 \xrightarrow{A} q_3 \end{array} \rightarrow \text{same group } G_2$$

$$\begin{array}{c} q_1 \xrightarrow{B} q_4 \\ q_2 \xrightarrow{B} q_5 \end{array} \rightarrow \text{same } G_2$$

$$\Pi_2 = \Pi_1$$



	a	b
q ₀	q ₁	q ₂
q ₁	q ₃	q ₄
q ₂	q ₃	q ₅
* q ₃	q ₃	q ₁
* q ₄	q ₄	q ₅
* q ₅	q ₅	q ₄
q ₆	q ₂	q ₆

1) first we get q₆ as unreachable from the starting point so it's removed

ii) group as (i) accepted states

{q₃, q₄, q₅} G₂

(ii) not accepted states

{q₀, q₁, q₂} G₁

$$\pi_0 = \{\{q_0, q_1, q_2\}, \{q_3, q_4, q_5\}\}$$

now divide further

Check for

q₀ q₁

* q₀, q₁ $\xrightarrow{A}$ q₁ q₃ diff group

* q₀, q₂ $\xrightarrow{A}$ q₂ q₅ "

* q₁, q₂ $\xrightarrow{A}$ q₃ q₃ } same group
 $\xrightarrow{B}$ q₄ q₅ }

so {q₀}, {q₁, q₂}

for {q₃, q₄, q₅}

* q₃, q₄ $\xrightarrow{B}$ q₁ q₅ different

* q₃, q₅ $\xrightarrow{B}$ q₁ q₄ "

so {q₃}, {q₄, q₅}

* q₄, q₅ $\xrightarrow{A}$ q₄ q₅ } same
 $\xrightarrow{B}$ q₅ q₄ }

$$\pi_1 = \{\{q_0\}, \{q_1, q_2\}, \{q_3\}, \{q_4, q_5\}\}$$

for π_2

q₁ q₂

q₁ q₂

$\xrightarrow{A}$

q₃ q₃

$\xrightarrow{B}$

q₄ q₅

Same group

q₄ q₅

$\xrightarrow{A}$

q₄ q₅

$\xrightarrow{B}$

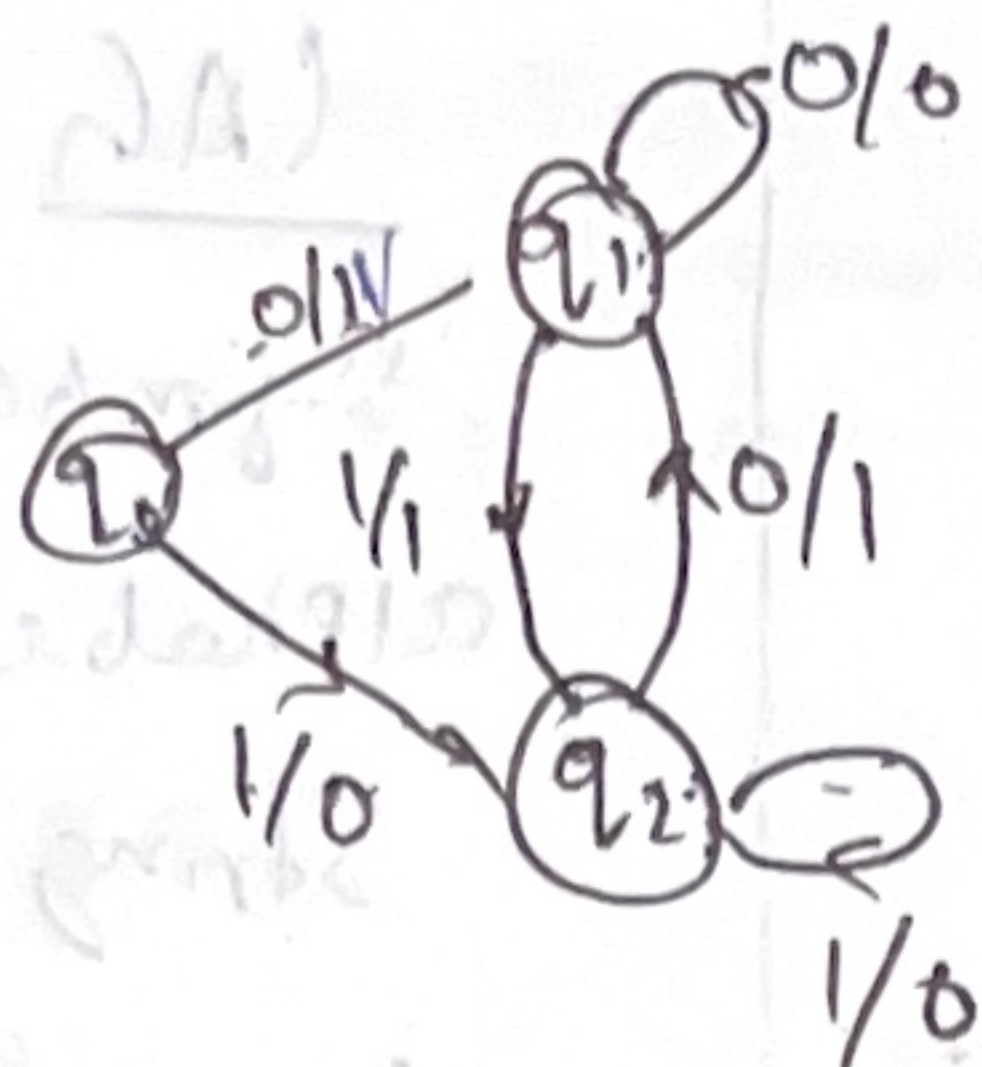
q₅ q₄

Same group

Same group

$$\pi_2 = \{\{q_0\}, \{q_1, q_2\}, \{q_3\}, \{q_4, q_5\}\}$$

1) 01010

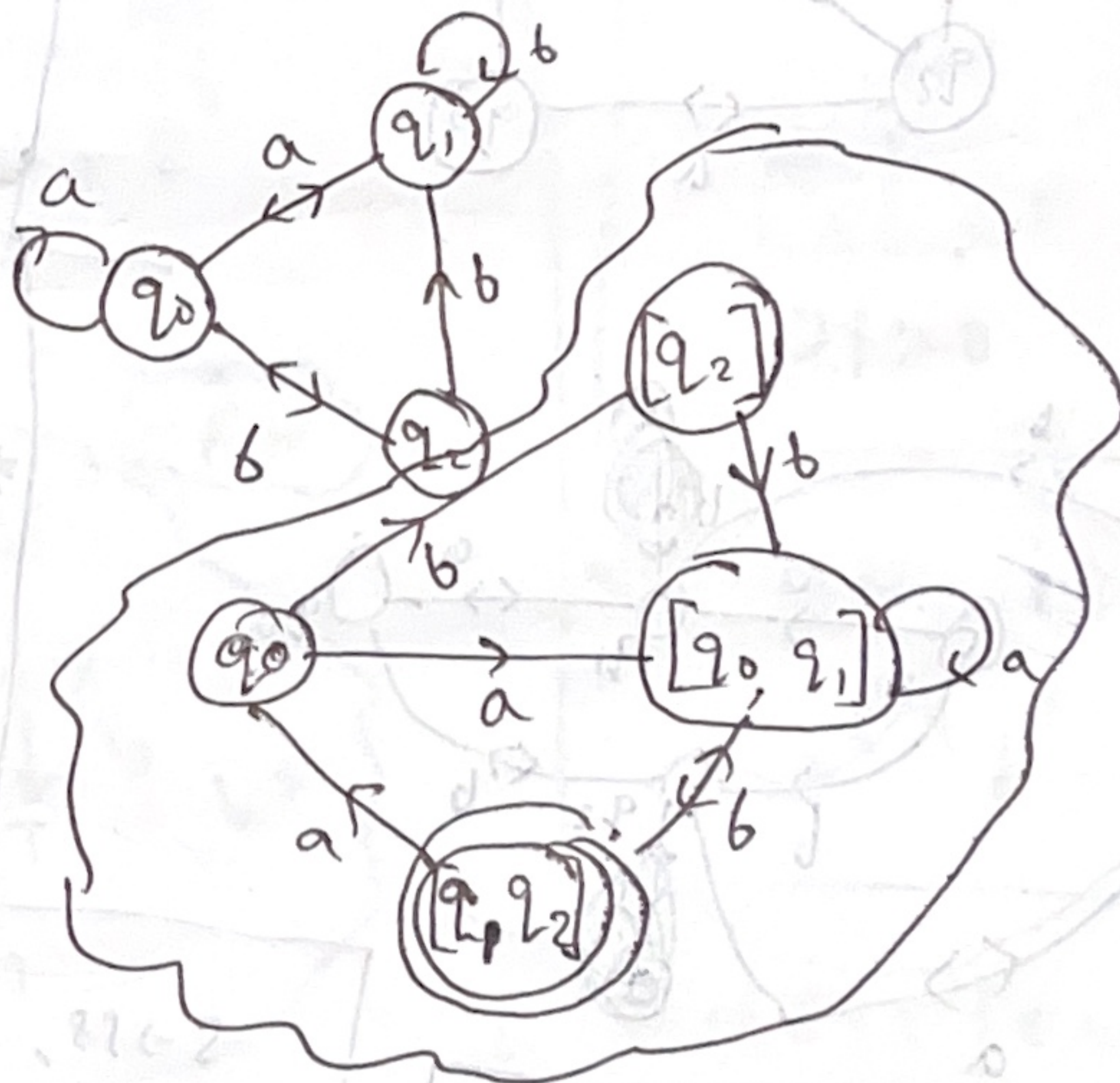
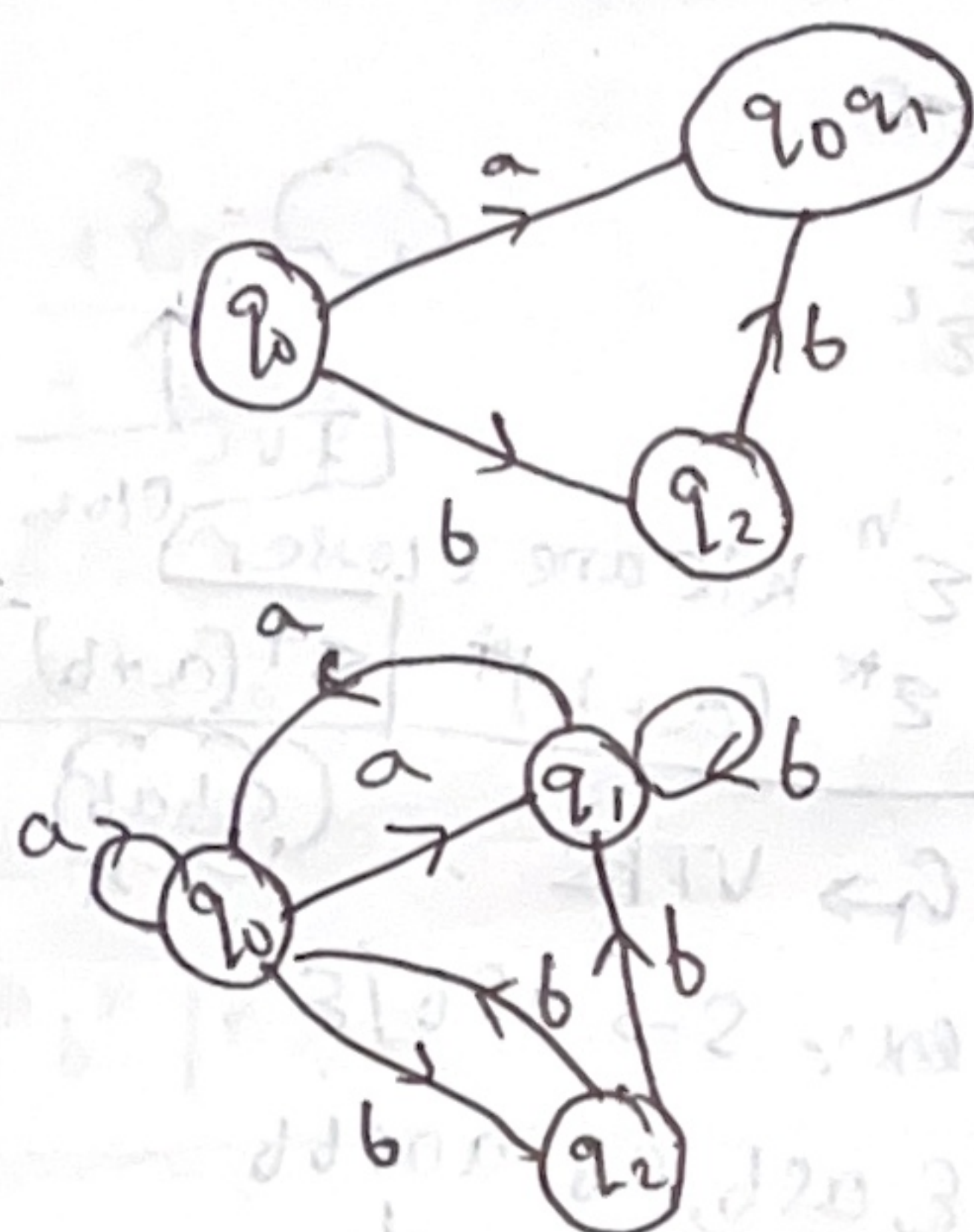


q ₁	q ₂	q ₁	q ₂	q ₁
1	1	1	1	1

2) NFA = $M = (\{q_0, q_1, q_2\}, \{a, b\}, \delta, q_0, \{q_2\})$

⑤

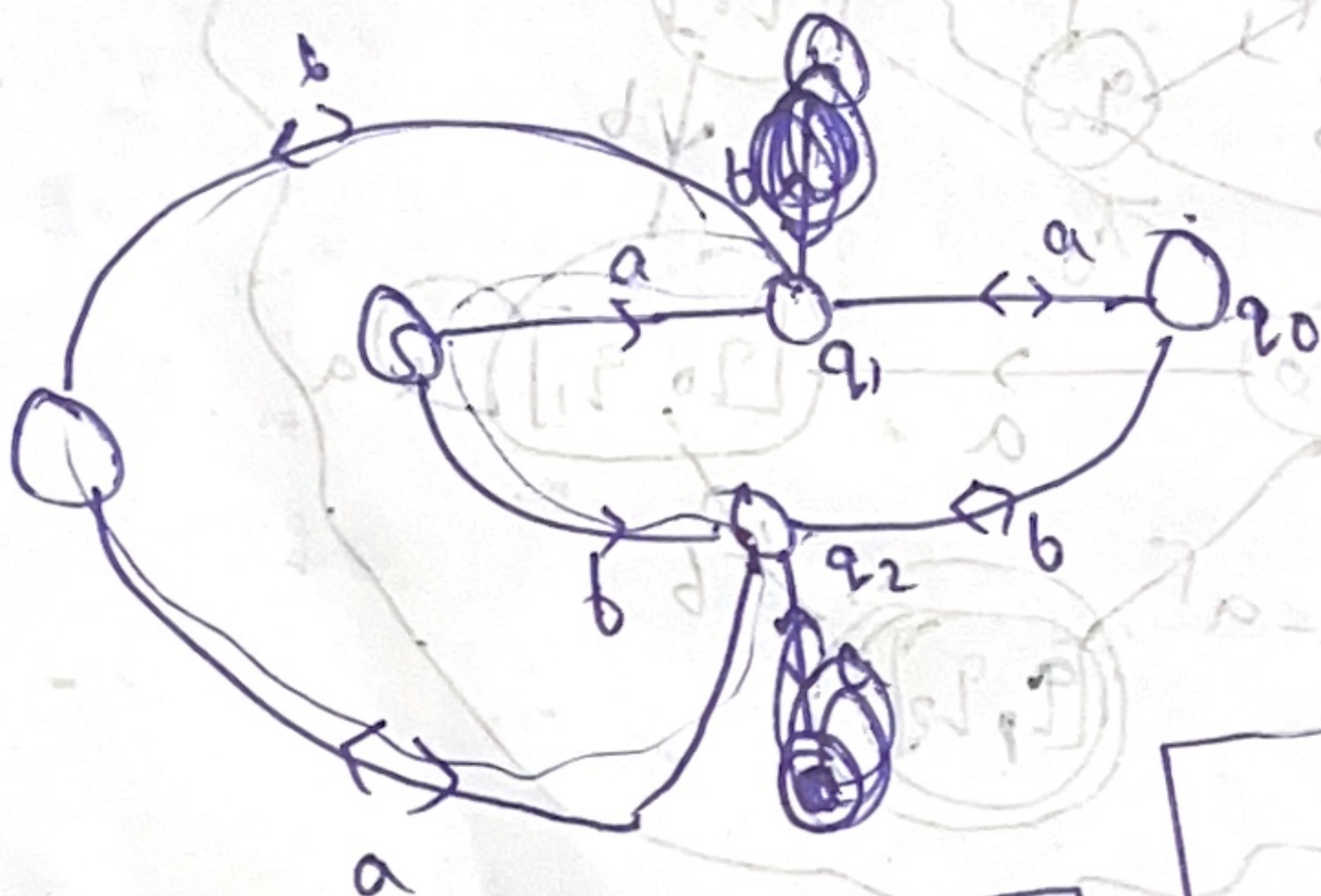
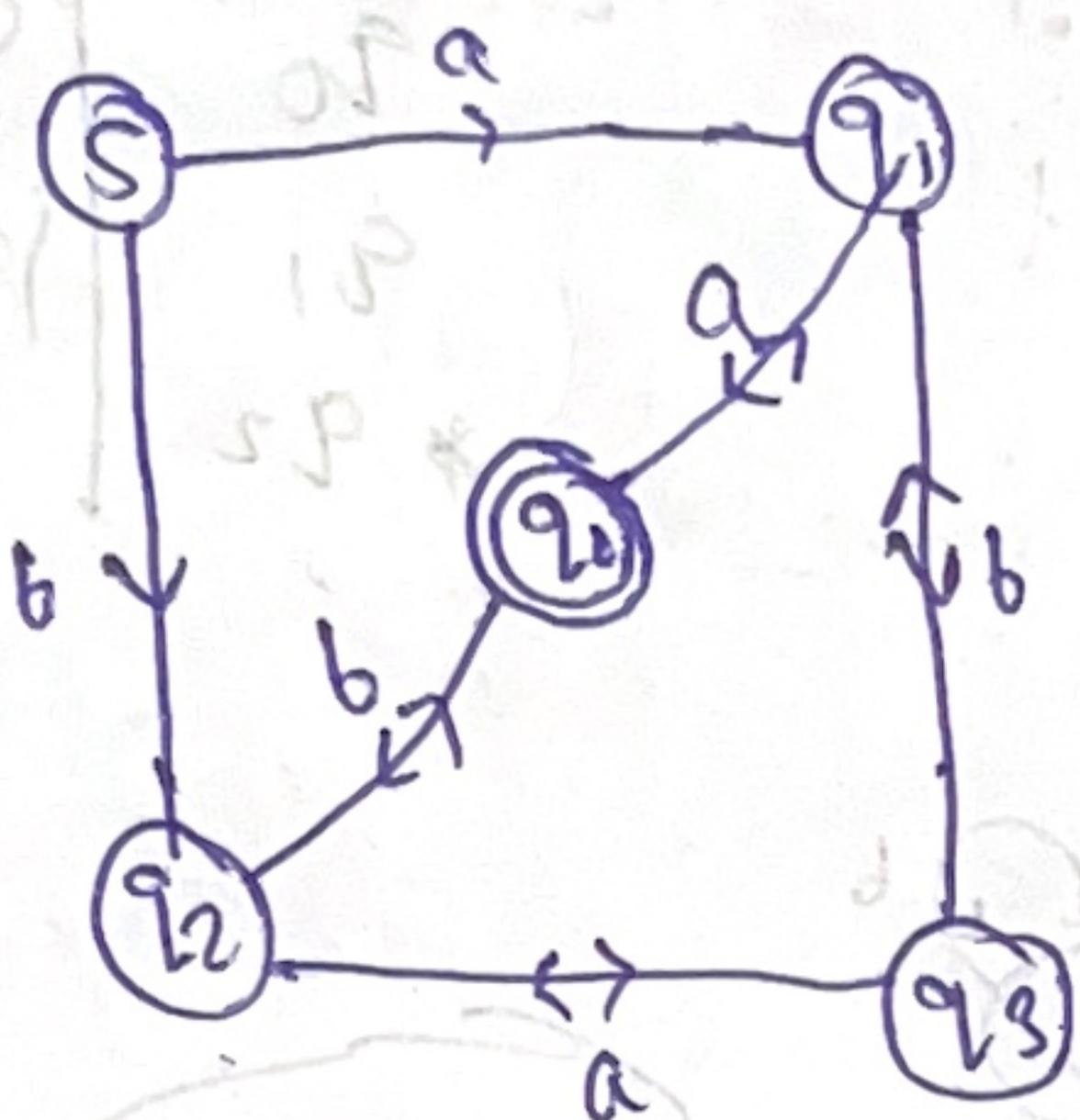
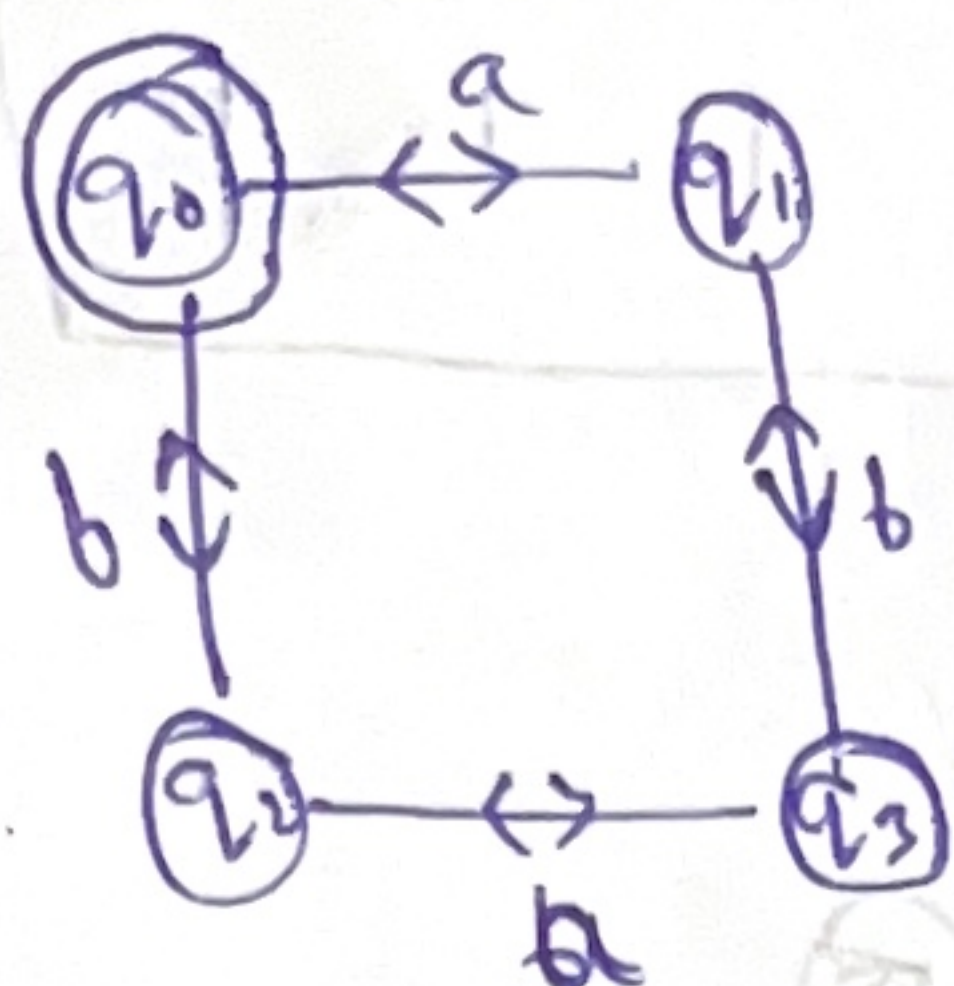
Current State	next state	
	a	b
q ₀	{q ₀ , q ₁ }	{q ₂ }
q ₁	{q ₀ }	{q ₁ }
q ₂	—	{q ₀ , q ₁ }



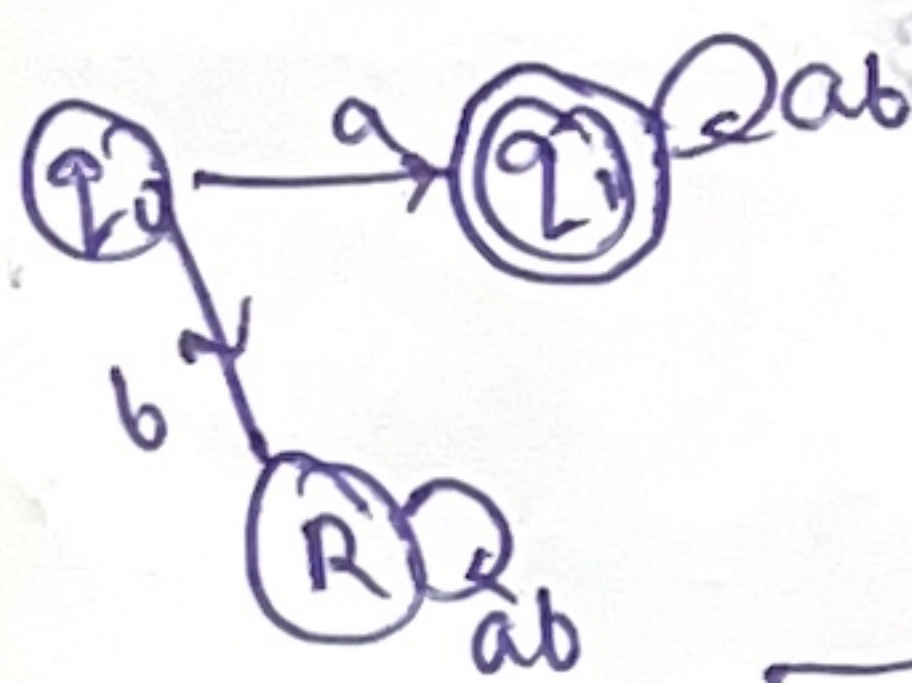
ab/a
 $\alpha \rightarrow \beta c/a$
 $\alpha \rightarrow \alpha d$

$A \rightarrow NUT^*$

$\alpha \neq \epsilon$
 $\alpha \neq \epsilon \rightarrow \alpha \neq \epsilon$



DFA $(Q, \Sigma, \delta, q_0, F)$



$\delta: Q \times \Sigma \rightarrow Q$
 $q_0 (a,b)$
 q_1
 R

$\{0a, 10b, 1a, 1b, Ra, Rb\}$

$q_0 q_1 R$

a...b



LAG

Symbol

alphabet Σ

String (len)

Language

memory / automata

Σ^0
 Σ^1
 Σ^2
 $\vdots$
 Σ^n
 Σ^*

Kleane (closure)

$\Sigma^+(a+b)$

$\Sigma^+(a+b)^*$

$\Sigma^+(a+b)^*$

$G \rightarrow VTP$

ex: $S \rightarrow a^nb^n / \epsilon$

$\epsilon, a^nb^n, ab, a^2b^2$

$aabb,$

$a^2b^2 \rightarrow a^2b^2$

$a^n b^n \quad n \geq 0$

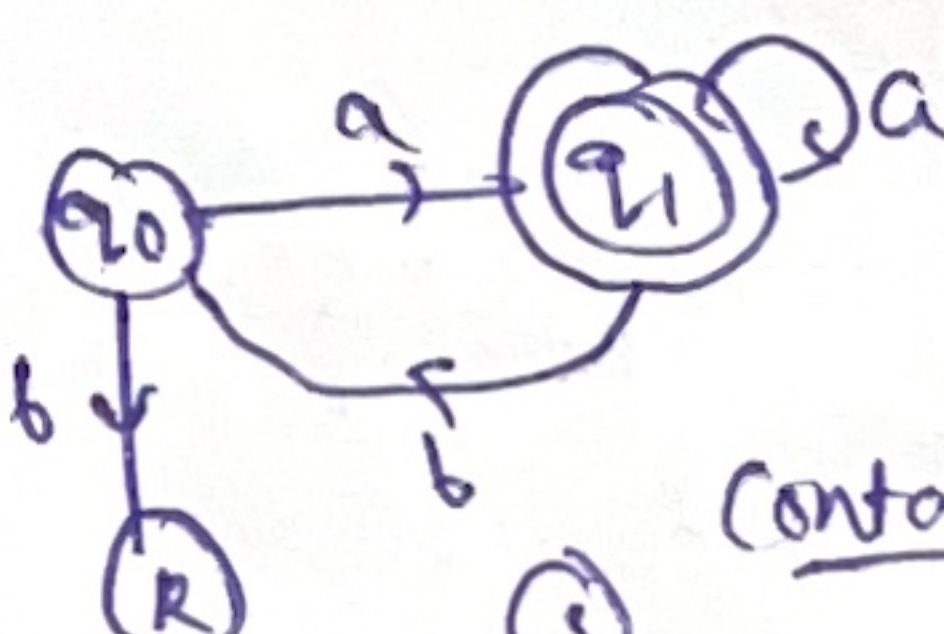
$S \rightarrow a^nb^n, S \rightarrow a^nb^n, S \rightarrow a^nb^n, S \rightarrow \epsilon$

$\epsilon, \frac{a^nb^n}{S} \frac{b^na^a}{S} \rightarrow abba$

No. of a = no. of b

$\begin{pmatrix} a \\ b \end{pmatrix}$

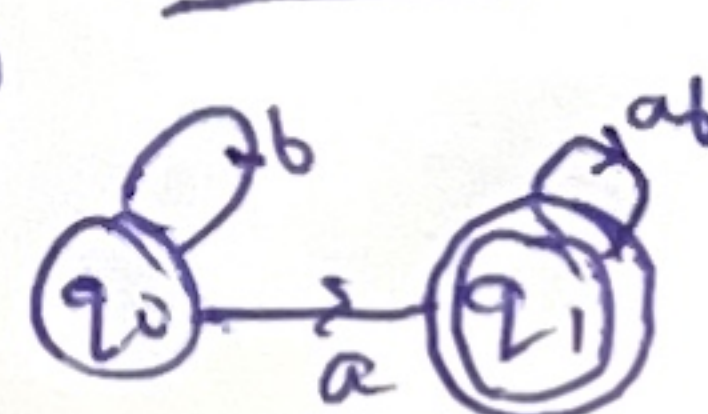
① Start with a end with a



② end with a



Contain a



$$G = (V, \Sigma, P, S)$$

$V \rightarrow$ Variable
 $\Sigma \rightarrow$ Terminal
 $P \rightarrow$ Production
 $S \rightarrow$ start

xyz
xy²zw²

CFG

PDA

$A \rightarrow d$

$FA + stack$

Regular lang

Context free lang.

$$M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$$

$$3) A \rightarrow AB$$

$$A \rightarrow a$$

2)

CFG

1 ~~non~~ v.

$$A \rightarrow a$$

$$S \rightarrow a^i b$$

$$S \rightarrow \epsilon$$

$$a^n b^n \mid n \geq 0$$

$$A \rightarrow a$$

 $\downarrow$
 any
 $\rightarrow a^i b$
 $\rightarrow \epsilon$

CFN

ENF

$$A \rightarrow BC$$

or

$$A \rightarrow a$$

$$\rightarrow 2|n| - 1$$

GNF

$$A \rightarrow a\alpha$$

(1)

$\downarrow$

T

V^+

n

$$\begin{aligned}
 &w \in \Sigma^* \\
 &w = xyz \\
 &|xy| \leq p \\
 &|y| > 0 \\
 &xy^i z \in L, i \geq 0 \\
 &a^{p-k} b^p
 \end{aligned}$$

$$G: E \rightarrow E + E \mid E * E \mid E = E \mid id$$

$$w: id + id + id$$

Adeny

$$q_0 \rightarrow aab + q_0 (atb)$$

$$R \rightarrow a + R(P)$$

$$R \rightarrow a^i b^j$$

$$= a^i b^j (a + b)^k$$

$$\begin{aligned}
 &a \rightarrow b \mid a \rightarrow b \\
 &a \rightarrow AB \mid a \rightarrow aab
 \end{aligned}$$

null pm

cmh

$|w| > p$

$w = xyz$

$|xy| \leq p$

$|y| > 0$

$xy^iz \in L, i \geq 0$

$x+yz$ must lie in the first p symbols $|xy| \leq p$

$y \rightarrow$ can't empty

$w = a^p b a^p \in L$

$|xy| \leq p$
 $|y| > 0$

$y = a^k, k > 0$

$i=0, xz, a^{p-k} b a^k$

$L = \{w \in \{a,b\}^* \mid w = w^R\}$

$\Sigma = \{a,b\}$

$\delta \rightarrow a s a \mid b s b \mid a \mid b \mid \epsilon$

Q E 8 2 0 F